QUEEN'S UNIVERSITY BELFAST

Thermodynamics of Point Defects

Mohamed Abdelilah Fadla

m.fadla@qub.ac.uk

Research fellow

School of Mathematics and Physics

Queen's University Belfast

Outline

QUEEN'S UNIVERSITY BELFAST

PART 1

Why Defects Matter

Thermodynamics of point defects

1. **Why defects matter**
   What defects control in real materials
2. **Classification and notation**
   Vacancies, interstitials, substitutions, complexes
3. **Thermodynamic quantities**
   Enthalpy, entropy and free energy
4. **Equilibrium concentration**
   How many defects a crystal actually contains

PART 2

Chemical Potentials and the Fermi Level

Connecting thermodynamics to real materials

1. **Chemical potentials**
   The cost of exchanging atoms with a reservoir
2. **Competing phases**
   What limits the allowed range of μ
3. **The Fermi level**
   The cost of exchanging electrons
4. **Worked example**
   Doping, compensation and carriers: p- vs n-type

About Me: My Path Through Defect Physics

QUEEN'S UNIVERSITY BELFAST

PhD

Laghouat University, Algeria

Defect tolerance in halide perovskites

e.g. $CsPbI_3$

Research fellow

CLMI, Queen's University Belfast

Defects in ultra-wide-bandgap semiconductors

e.g. Ga_2O_3, Al_2O_3

Research fellow

CQMT, Queen's University Belfast

Ferroelectric perovskites

$PbTiO_3$, $BaTiO_3$

The same question in every system: which defects form, and how do they change the material?





Where This Lecture Fits in the MAMBA School

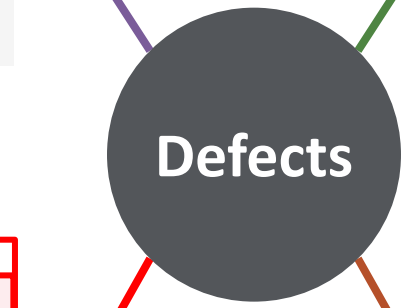
QUEEN'S UNIVERSITY BELFAST

EXPERIMENTS AND APPLICATIONS

- Experimental tools & defect metastability *(Tue 20)*
- Seeing defects with TEM *(Tue 20)*
- Space applications *(Tue 27)*
- Radiation detectors · Dosimetry *(Wed 28 · Fri 30)*

MODELLING ACROSS SCALES

- Electronic structure: QE & CP2K *(Mon 19 – Tue 20)*
- Molecular dynamics with LAMMPS *(Thu 22)*
- Machine-learning force fields *(Mon 26 – Tue 27)*



TODAY · You are here

DEFECT FORMATION AND CHARGE

- **Thermodynamics of point defects** *(09:00)*
- **Formation energies & transition levels** *(11:00)*
- Defect structure & spectroscopy *(Thu 22)*
- Point-defect calculations with QE *(Thu 22)*

DEFECT MOTION AND EVOLUTION

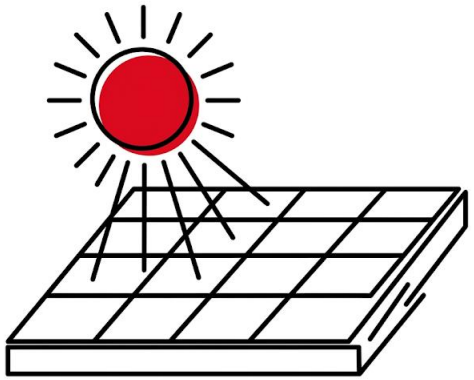
- Transition pathways: NEB & ART *(Wed 21, next!)*
- Cascades with MD *(Thu 22)*
- Defect evolution & annealing (KMC) *(Fri 23)*
- Defects in irradiated materials *(Mon 26)*

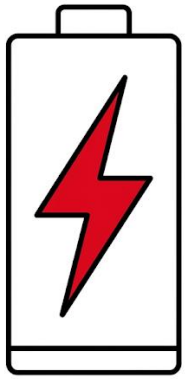
Motivation and Importance of Defects

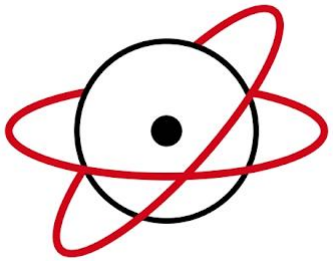
QUEEN'S UNIVERSITY BELFAST

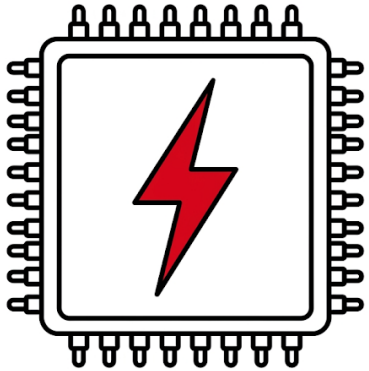
- Many physical properties of functional materials are governed by defects, not bulk characteristics
- Defects activate electronic and ionic conductivity
- Defects create active centres for catalysis
- Defects store information through localised spin configurations

Applications: Photovoltaics · Batteries · Power electronics · Quantum computing . Radiotherapy



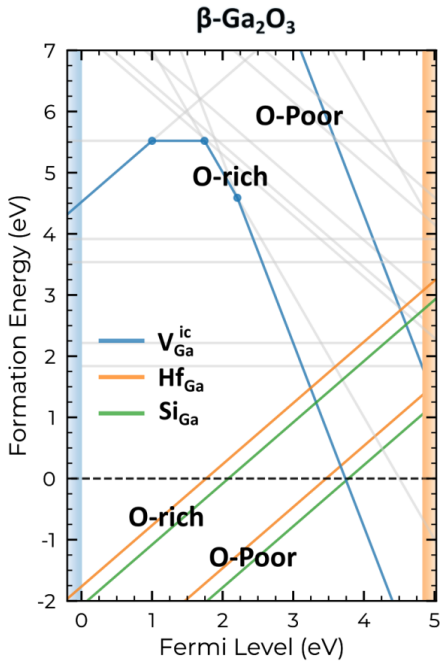




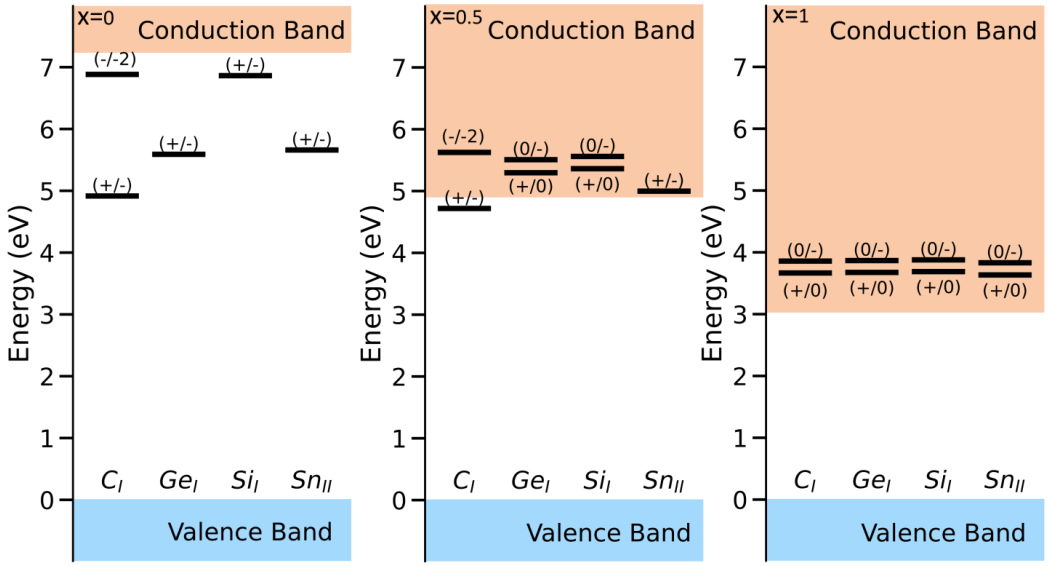


By the End of These Lectures, You Can Calculate…

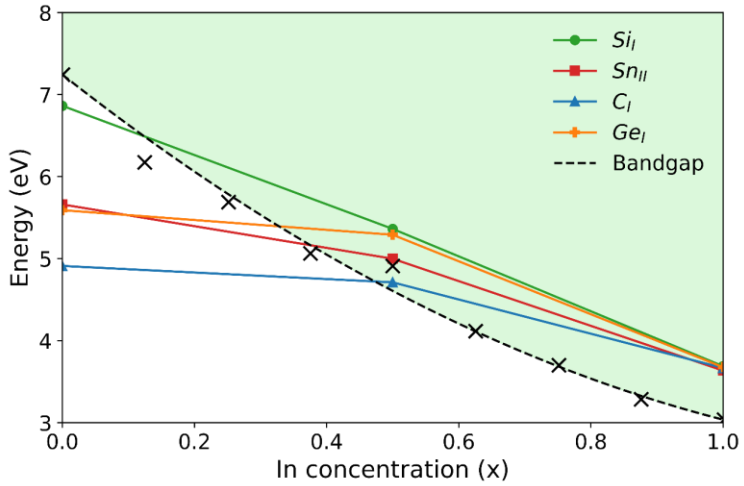
QUEEN'S UNIVERSITY BELFAST



Defect formation energies



Thermodynamic transition levels



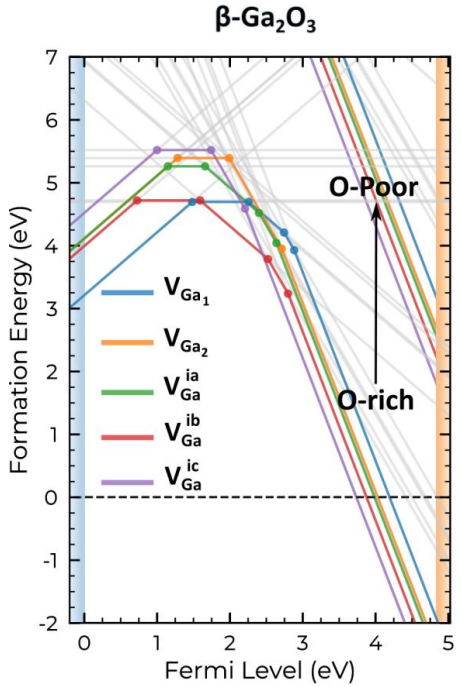
Critical composition

Fadla et al., arXiv:2605.28743 (2026)

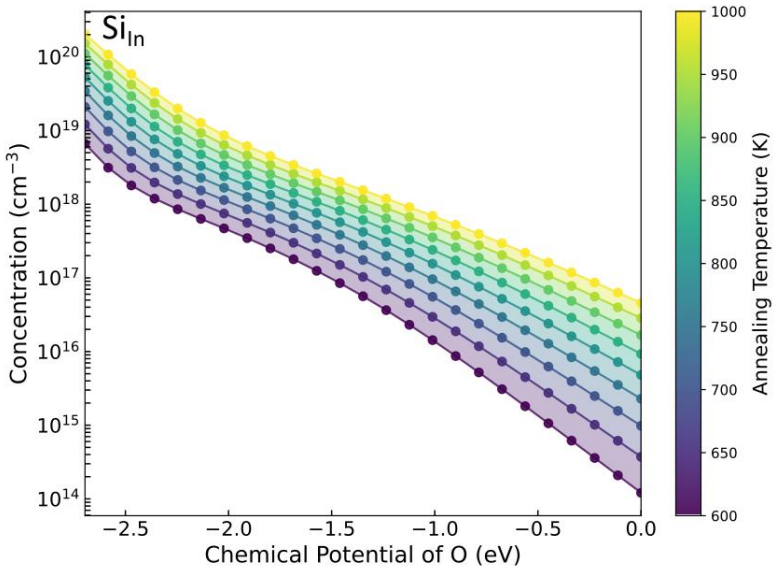
Fadla *et al.*, *PRMaterials* **9**, 105002 (2025)

By the End of These Lectures, You Can Calculate…

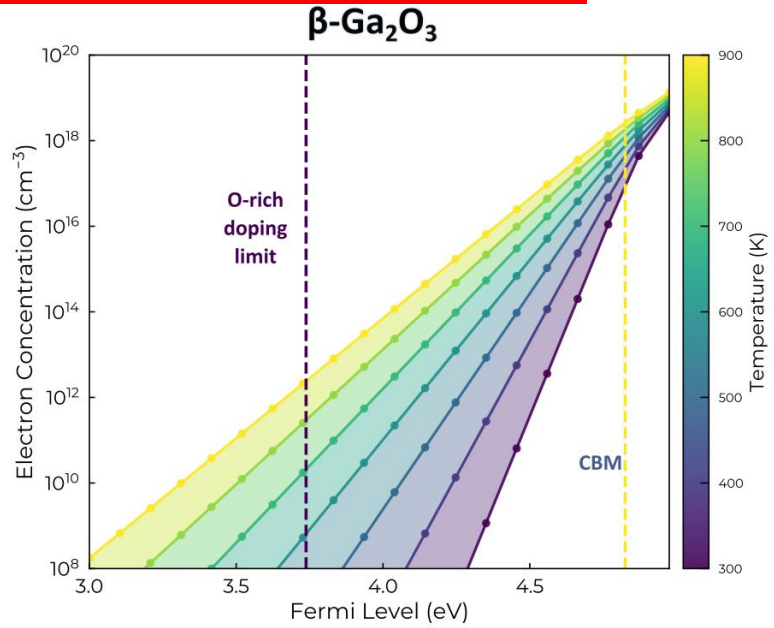
These lectures treat defects in thermal equilibrium. Under irradiation, defects are produced far from equilibrium. What I give you, formation energies and charge states … , are the key inputs for those models.



Defect formation energies



Defect density



Carrier concentration

Fadla et al., arXiv:2605.28743 (2026)

Fadla *et al.*, *PRMaterials* **9**, 105002 (2025)

Classification and Terminology of Point Defects

QUEEN'S UNIVERSITY BELFAST

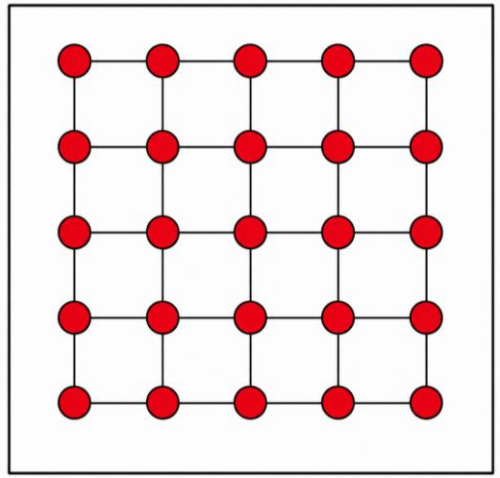
- Intrinsic vs. Extrinsic defects:

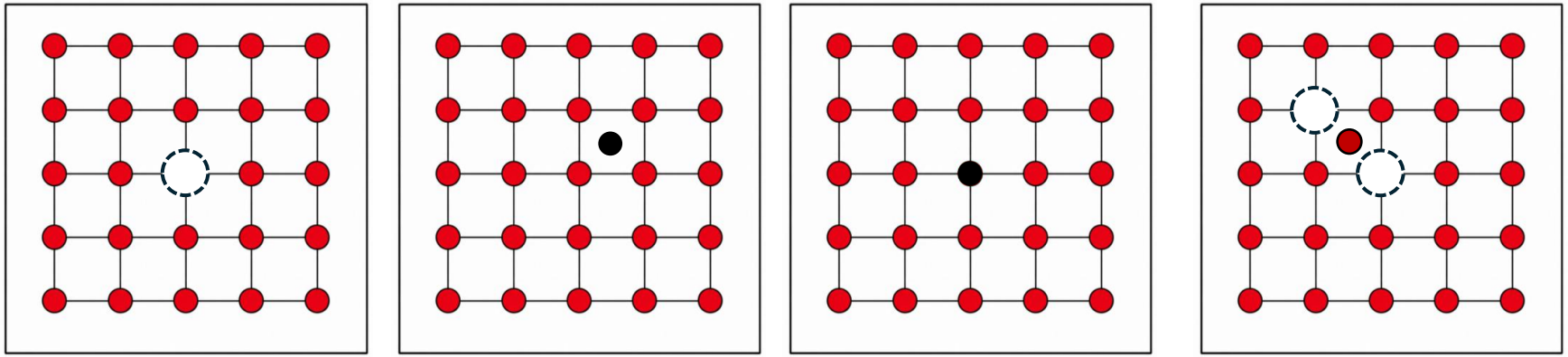
Imperfections from bulk atoms vs. external impurity doping.

- Types of point defects: Vacancies, interstitials, and substitutions (anti-sites).

Defect Complexes: Combinations of multiple interacting point defects. (Several ways of interactions)

Pristine





Vacancy | Interstitial | Substitution | Split vacancy

Note: Extended defects, such as grain boundaries and interfaces, can affect charge redistribution, recombination, and local band alignment.

Classification and Terminology of Point Defects

QUEEN'S UNIVERSITY BELFAST

Notation:

Elemental species, Lattice sites, Relative electronic charge states.

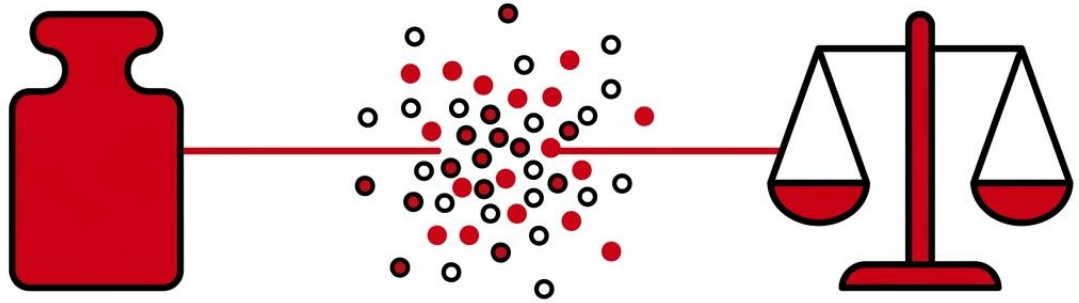
Defect	Defect-physics notation	Species	Site	Effective charge
Oxygen vacancy	V_O^{2+}	O	O	+2
Ga interstitial	$\mathrm{Ga_i^{3+}}$	Ga	i	+3
Mg on Ga site	$\mathrm{Mg_{Ga}^{-}}$	Mg	Ga	−1

Thermodynamic Quantities

QUEEN'S UNIVERSITY BELFAST

- **Enthalpy:** creating the defect costs energy

- **Entropy:** Many ways to place the defects; each defect can also change vibrations, spin and orientation.



- **Free Energy:** combines enthalpy and entropy to determine what the crystal actually does

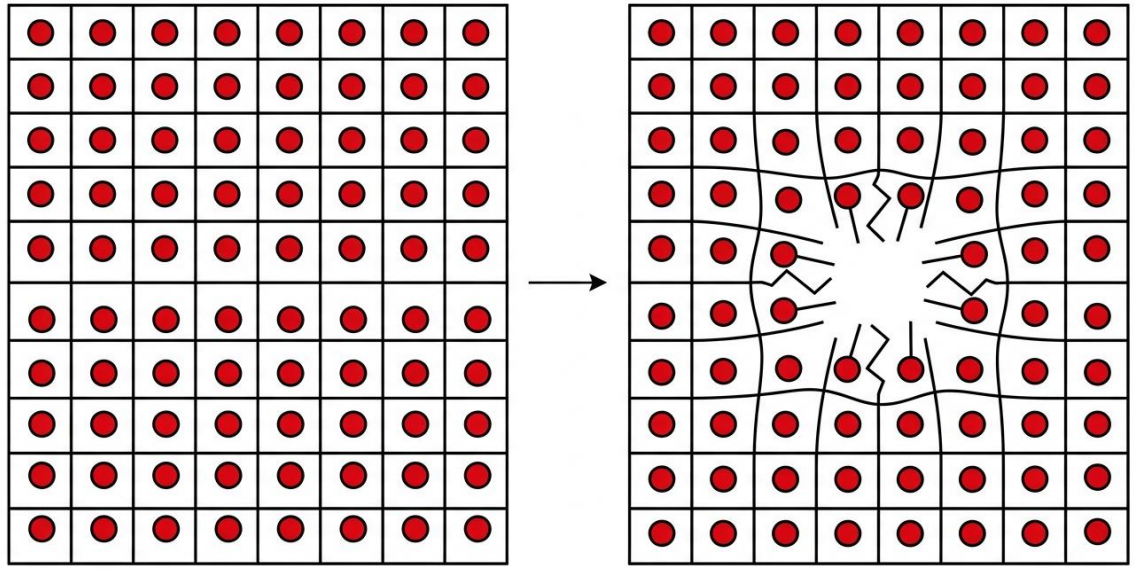
Defect concentration is ultimately a competition between:
Energetic cost vs Entropic benefit

Enthalpy H: How much energy does the process cost?

QUEEN'S UNIVERSITY BELFAST

For a defect reaction:

Perfect crystal → crystal + defect + atom(s) to a reservoir



Creating the defect requires breaking bonds, distorting the lattice, etc., that costs energy.

$$\Delta H_f > 0$$

It takes energetic effort to create this defect.

For solids at ordinary pressures, the distinction between internal energy and enthalpy is often relatively small, because the PV contribution is tiny.

$$H \approx E$$

Is it possible to get $\Delta H_f < 0$?!

In equilibrium, $\Delta H_f < 0$ usually signals a setup error (missing competing phases, μ outside the stability region, bad corrections)

Why can defects exist even though they cost energy?

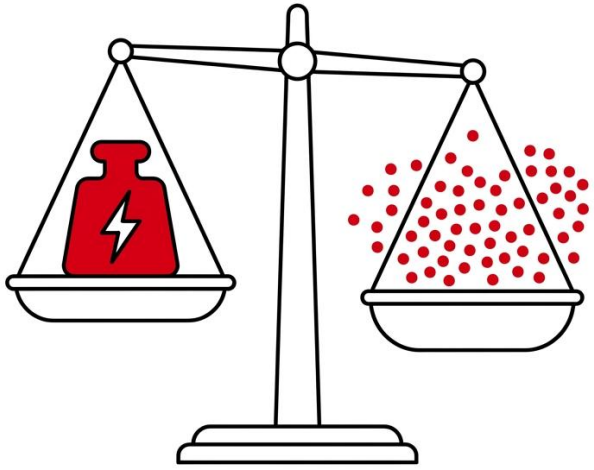
QUEEN'S UNIVERSITY BELFAST

Suppose a vacancy costs

$$\Delta H_f = 2\ eV.$$

The equilibrium crystal should have zero vacancies?!

energy penalty ΔH_f – entropy reward $T\Delta S$



Creating vacancies gives the crystal additional configurational possibilities.
The entropy contribution $-T\Delta S_{conf}$ lowers the free energy.

The equilibrium crystal therefore contains **some finite concentration of defects**, even if every individual defect costs energy.

Entropy S: How many ways can the system be arranged?

QUEEN'S UNIVERSITY BELFAST

❑ **For defects**

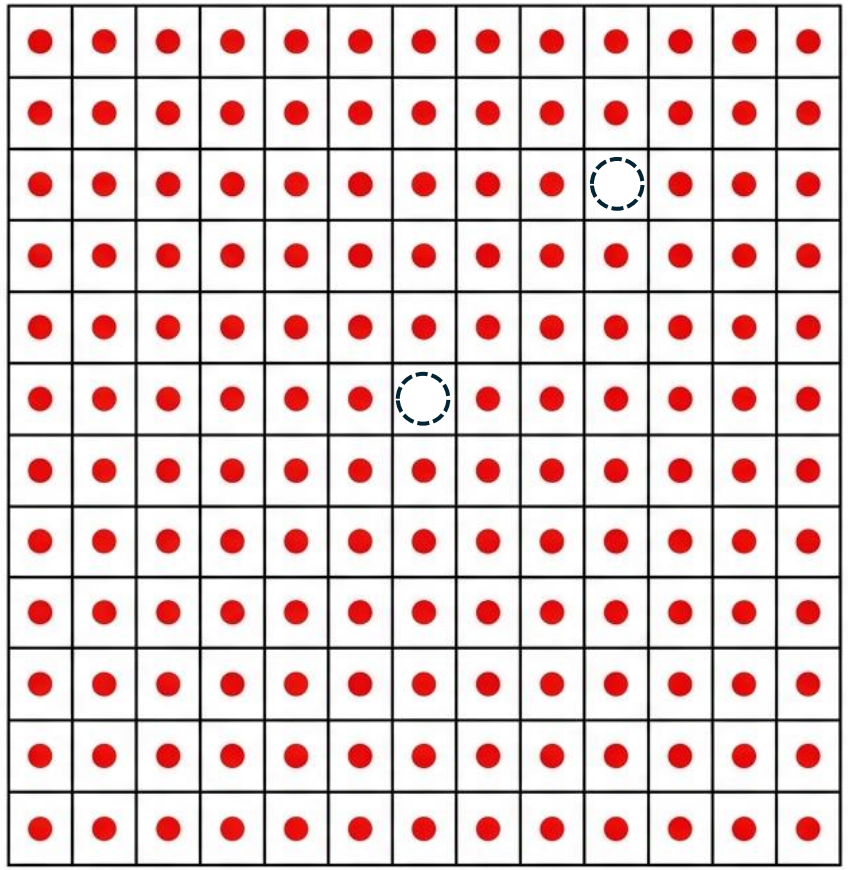
Entropy measures how many **microscopic** configurations are compatible with the **macroscopic** state.

The key idea:
More possible arrangements⇒larger entropy

Temperature determines how important this entropy is.
At T=0, energy dominates.

$$G = H - TS$$

At higher T, entropy becomes increasingly valuable.



Entropy S: How many ways can the system be arranged?

QUEEN'S UNIVERSITY BELFAST

❑ **For defects**

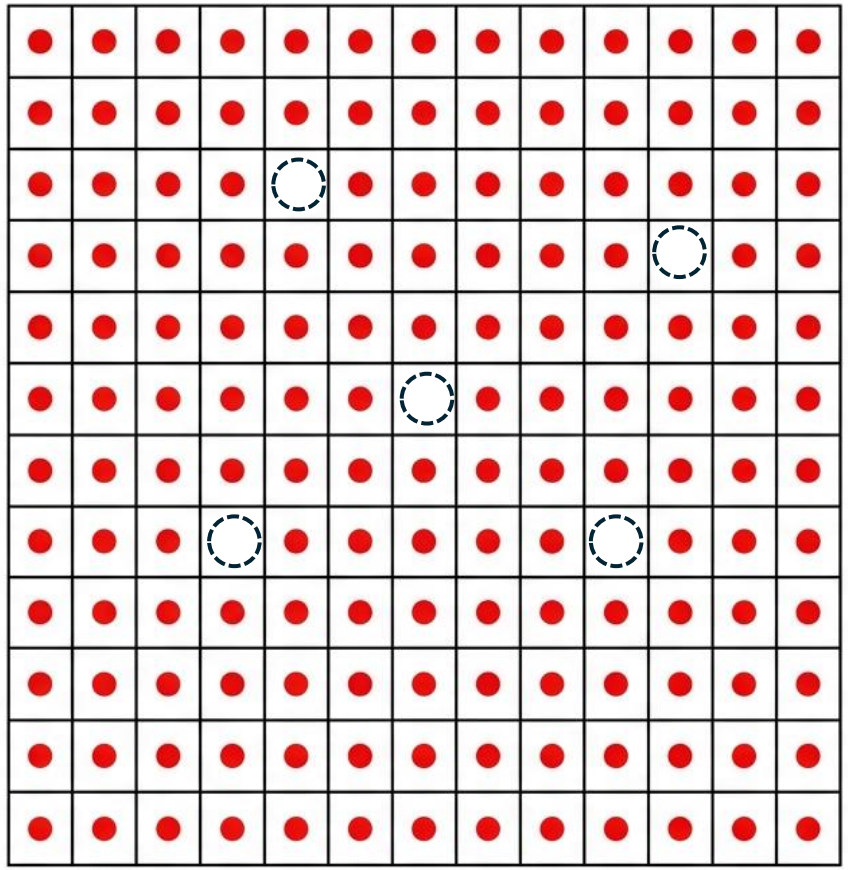
Entropy measures how many **microscopic** configurations are compatible with the **macroscopic** state.

The key idea:
More possible arrangements⇒larger entropy

Temperature determines how important this entropy is.
At T=0, energy dominates.

$$G = H - TS$$

At higher T, entropy becomes increasingly valuable.



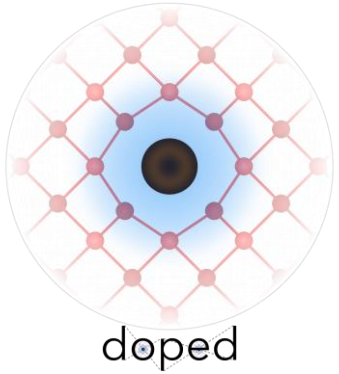
Entropy S: How many ways can the system be arranged?

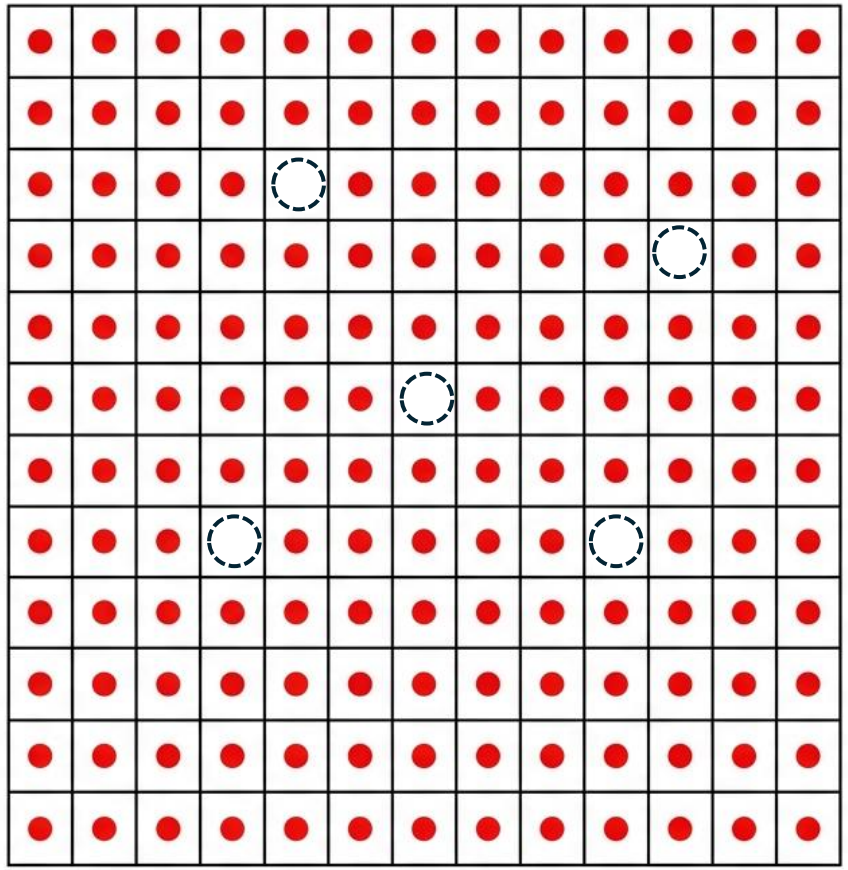
QUEEN'S UNIVERSITY BELFAST

Non-Configurational Entropy:

$$\Delta S_f = s_{orientational} + s_{spin} + s_{vibrational} + s_{electronic}$$

Defect	q	Site Symmetry	Defect Symmetry	g_{Orient}	g_{Spin}	g_{Total}	Mult
V_{Cd}	0	Td	C2v	6.0	1.0	6.0	1.0
	-1	Td	C3v	4.0	2.0	8.0	1.0
	-2	Td	Td	1.0	1.0	1.0	1.0





Check: doped.readthedocs.io

Free energy G: What does the system actually want?

QUEEN'S UNIVERSITY BELFAST

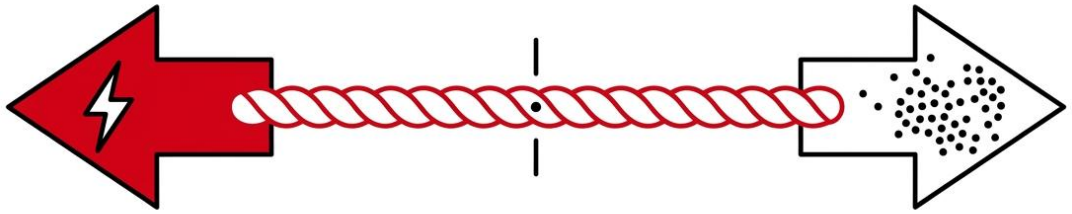
At constant temperature and pressure, the equilibrium state minimizes the **Gibbs free energy**:

$$G = H - TS$$

For defect formation:

$$\Delta g_f = \Delta h_f - T\Delta S_f$$

Energy says:
Don't make the defect; it costs Δh_f.
Configurational entropy says:
Make defects; there are many ways of arranging them.
Free energy says:
Let's combine both effects and see what wins.



F**ree energy**, rather than formation energy alone, determines equilibrium defect concentrations.

Equilibrium Concentration

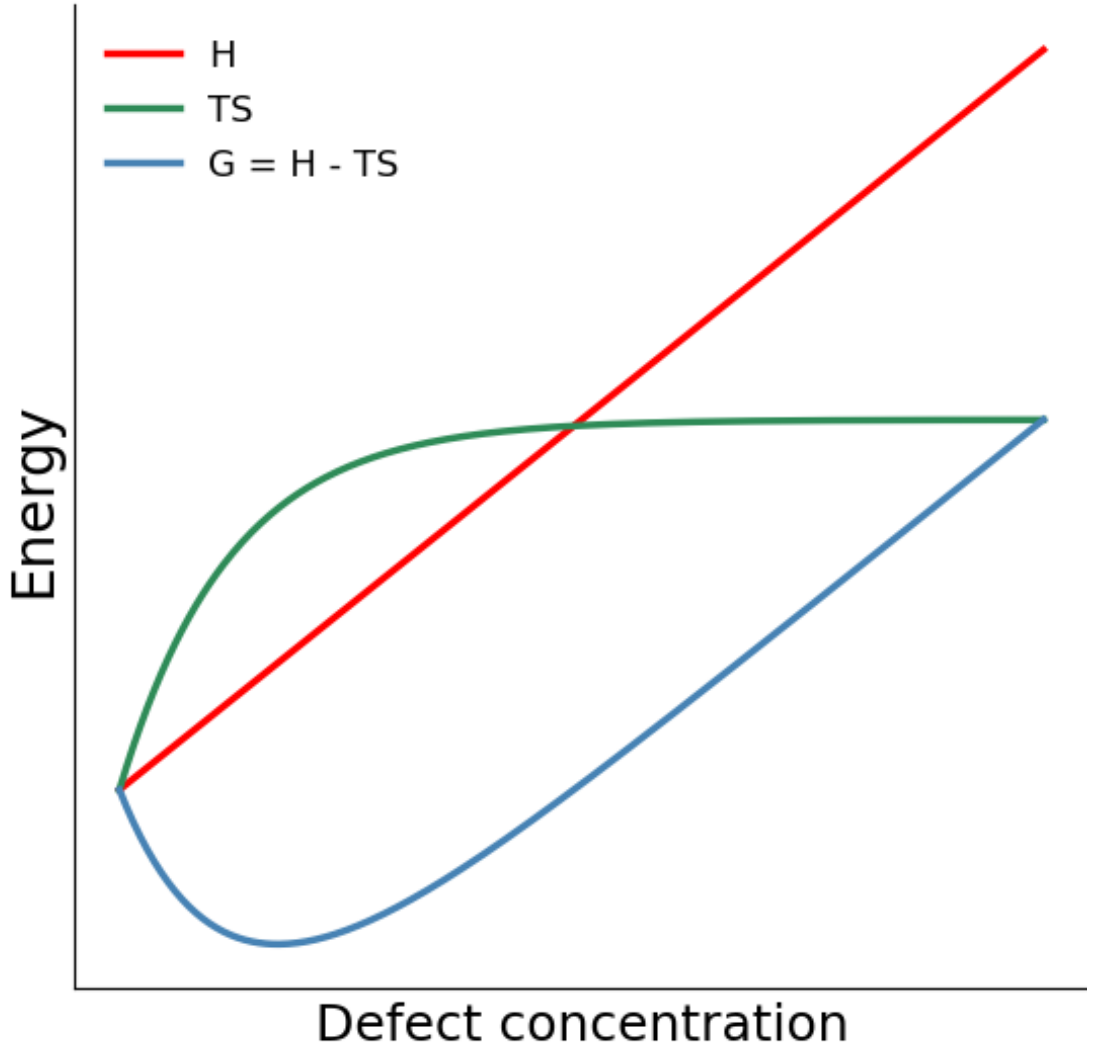
QUEEN'S UNIVERSITY BELFAST

$$\frac{\partial G_{d,P}}{\partial n_d} = \frac{\partial (n_d g_{f,P} - TS^{conf})}{\partial n_d} = 0$$

$$g_f = h_f - TS_{f,non-conf}$$

$$c_{eq} = \exp\left(-\frac{g_f}{k_B T}\right)$$

"site fraction"



Full demonstration: Irea Mosquera-Lois et al. Chem. Soc. Rev., 2023, 52, 5812

Equilibrium Concentration

QUEEN'S UNIVERSITY BELFAST

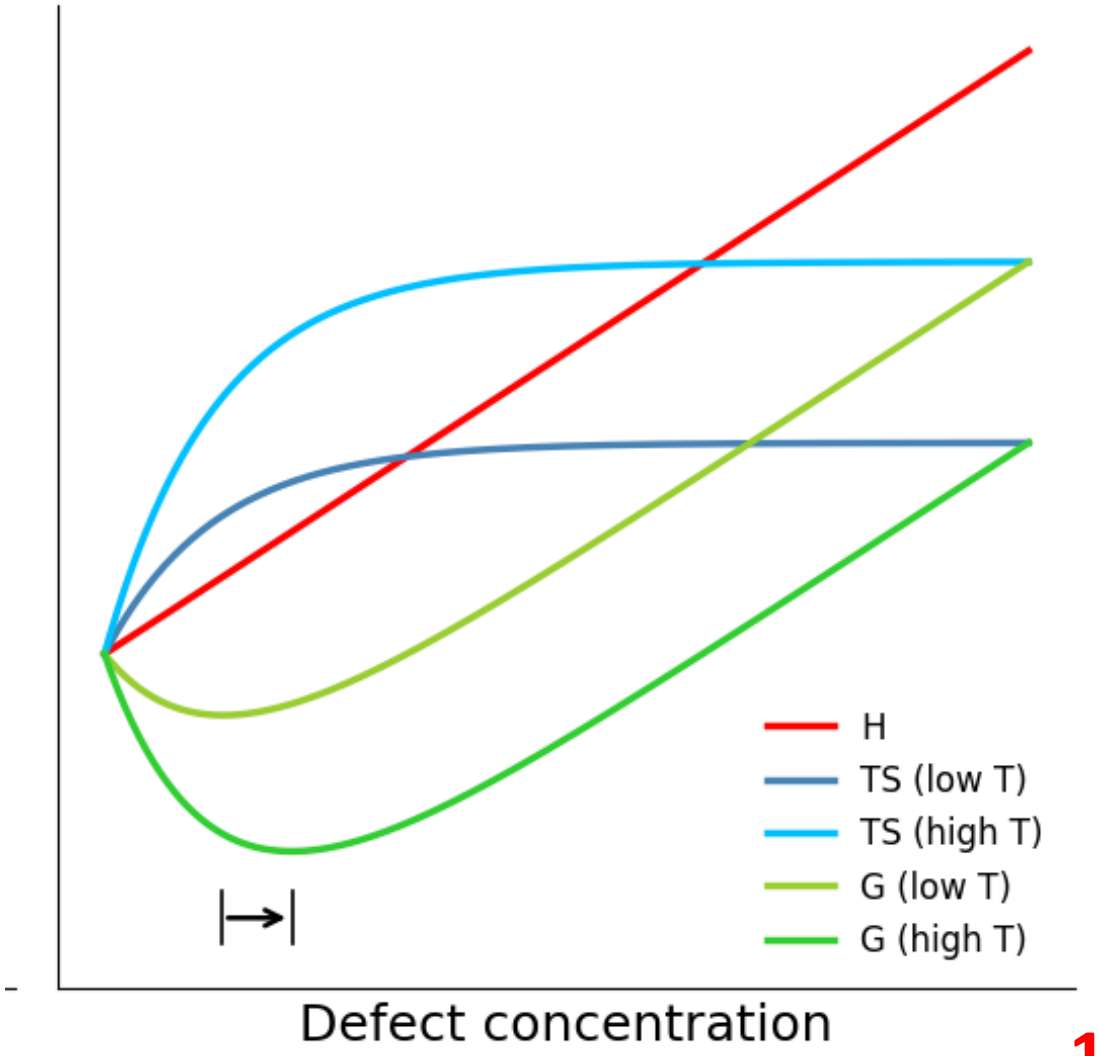
$$\frac{\partial G_{d,P}}{\partial n_d} = \frac{\partial (n_d g_{f,P} - TS^{conf})}{\partial n_d} = 0$$

$$g_f = h_f - TS_{f,non-conf}$$

$$c_{eq} = \exp\left(-\frac{g_f}{k_B T}\right)$$

"site fraction"

Higher T: the minimum moves to higher c.



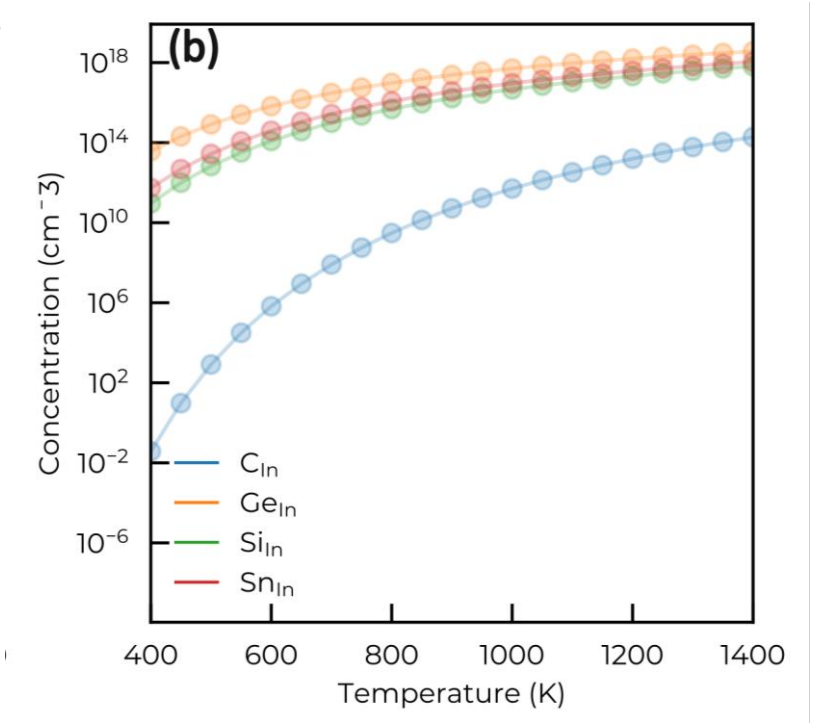
Full demonstration: Irea Mosquera-Lois et al. Chem. Soc. Rev., 2023, 52, 5812

What controls the concentration?

QUEEN'S UNIVERSITY BELFAST

$$c_{eq} = (N_{site}/V) \cdot g_{deg} \cdot \exp\left(-\frac{\Delta E_f}{k_B T}\right)$$ "per volume (cm⁻³)"

- $\boldsymbol{\Delta E_f}$: formation energy. Depends on μ_i and E_F (Part 2) and on DFT accuracy (Lecture 3).
- **g:** degeneracy and entropy terms, $\exp(\Delta s_f/k_B)$ (this part).
- **T:** growth or anneal temperature vs operating temperature (concentrations frozen in).
- $\boldsymbol{N_{sites}/V}$**:** number of available sites per volume (site multiplicity).



Fadla *et al.*, *PRMaterials* **9**, 105002 (2025)

Exponential in energy, so small errors give large changes in concentration.

- High Δh_f → fewer defects
- High Δs_f → more defects
- High T → entropy matters more
- Chemical environment → μ_i shifts ΔE_f (next part)

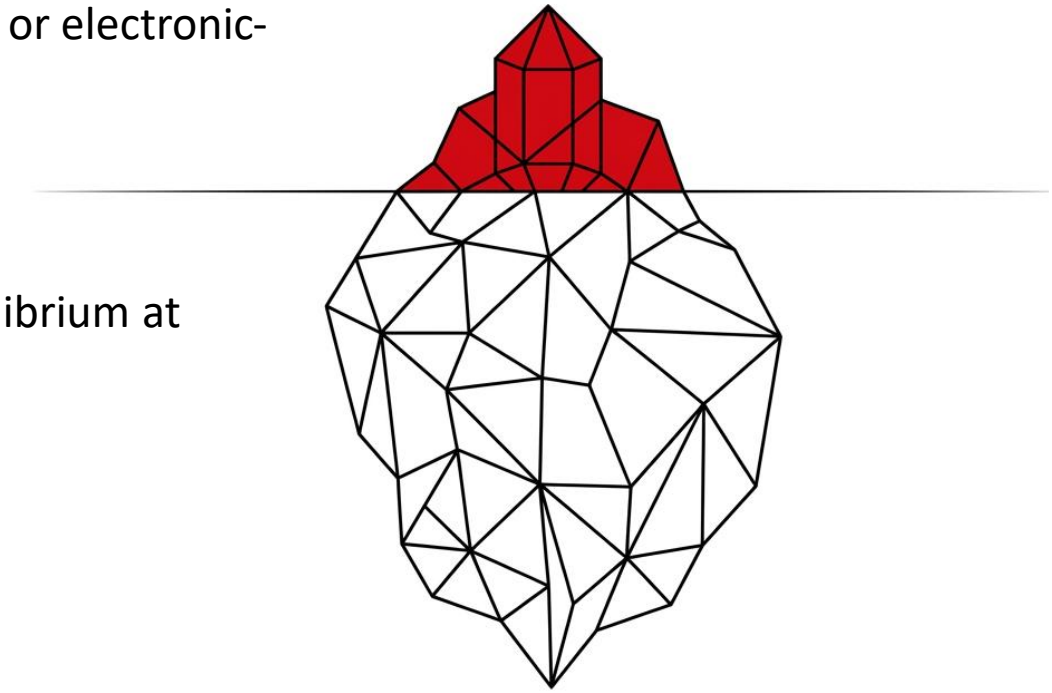
Formation Free Energy ≠ Formation Energy

QUEEN'S UNIVERSITY BELFAST

- **Formation energy**
  → primarily an energetic quantity, often the T=0 or electronic-structure quantity obtained from DFT.

- **Formation free energy**
  → the thermodynamic quantity relevant to equilibrium at finite T:

$$\Delta G_f(T) = \Delta H_f(T) - T\Delta S_f(T).$$



DFT formation energies should not be treated as if they were the whole story!

Key Takeaways

QUEEN'S UNIVERSITY BELFAST

Quantity	Question
Enthalpy H	How much energetic cost is involved?
Entropy S	How many microscopic ways can I arrange things?
Free energy G	After balancing energy and entropy, what state is preferred and how many defects do we have?

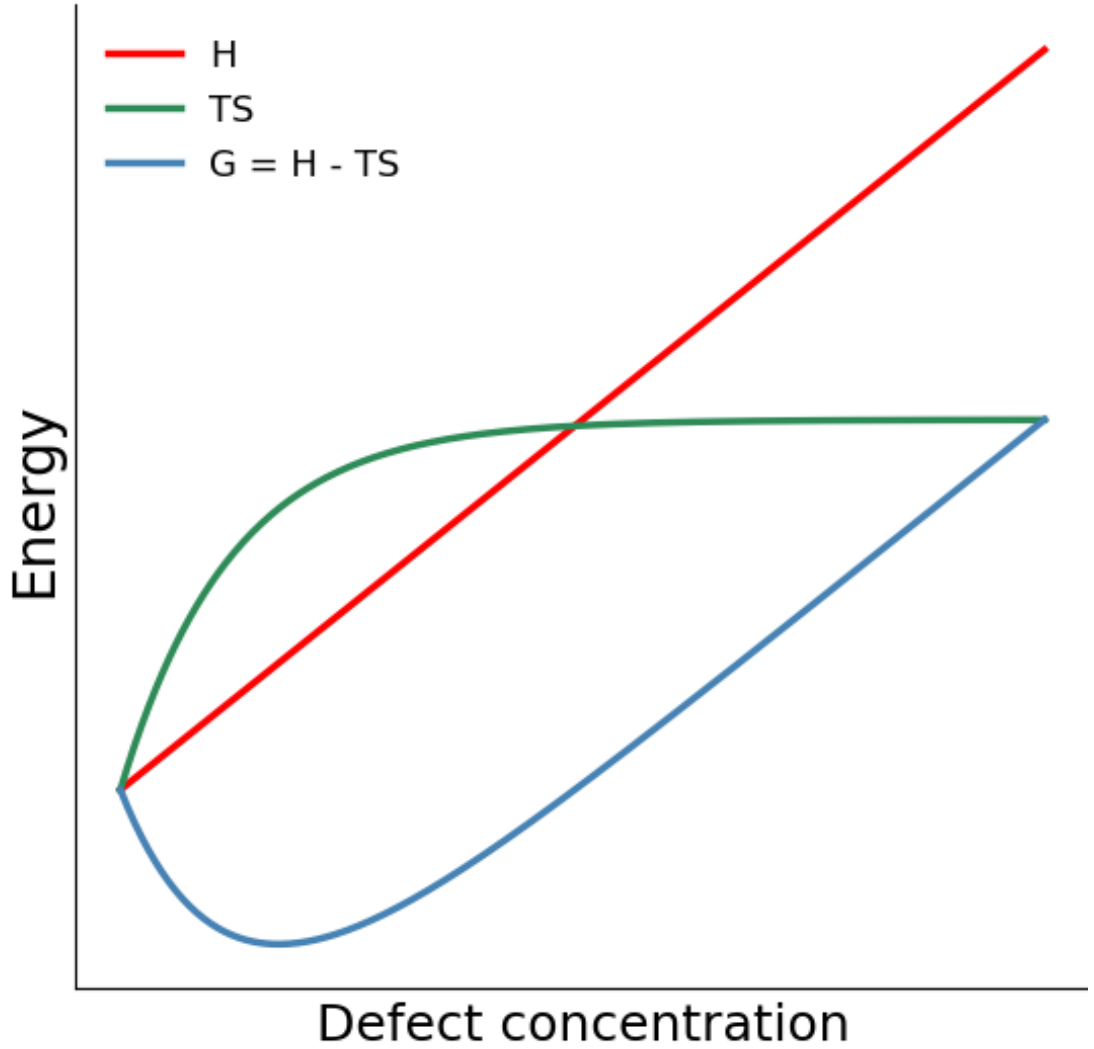
Part 2: Chemical Potentials and the Fermi Level — Connecting Thermodynamics to Real Materials

Recap: What We Covered

QUEEN'S UNIVERSITY BELFAST

- **Enthalpy H**
- **Entropy S**
- **Free energy G**

$$c_{eq} = \exp\left(-\frac{\Delta g_f}{k_B T}\right)$$



Outline

QUEEN'S UNIVERSITY BELFAST

PART 1

Why Defects Matter

Thermodynamics of point defects

1. **Why defects matter**
   What defects control in real materials
2. **Classification and notation**
   Vacancies, interstitials, substitutions, complexes
3. **Thermodynamic quantities**
   Enthalpy, entropy and free energy
4. **Equilibrium concentration**
   How many defects a crystal actually contains

PART 2

Chemical Potentials and the Fermi Level

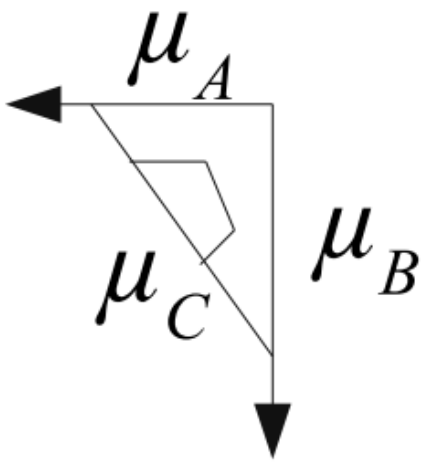
Connecting thermodynamics to real materials

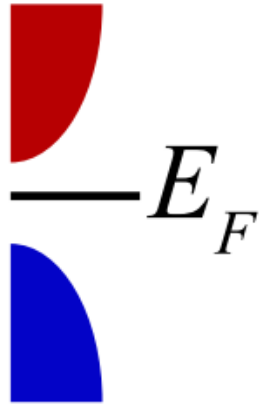
1. **Chemical potentials**
   The cost of exchanging atoms with a reservoir
2. **Competing phases**
   What limits the allowed range of μ
3. **The Fermi level**
   The cost of exchanging electrons
4. **Worked example**
   Doping, compensation and carriers: p- vs n-type

Where μ_i and E_F enter

$$\Delta E_f(X^q) = E[X^q] - E[bulk] - \boldsymbol{\Sigma\, n_i \mu_i} + \boldsymbol{q(E_{VBM} + E_F)} + E_{corr}$$

Now: the two terms in bold. Next lecture: everything else





Atom reservoir
μ_i (growth conditions)

⇄

Crystal + defect
$\Delta H_f(D, q)$

⇄

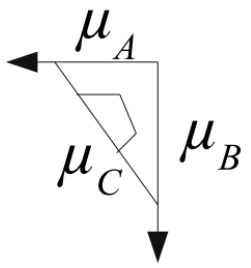
Electron reservoir
E_F (doping, carriers)

Chemical Potentials and Competing Phases (intro)

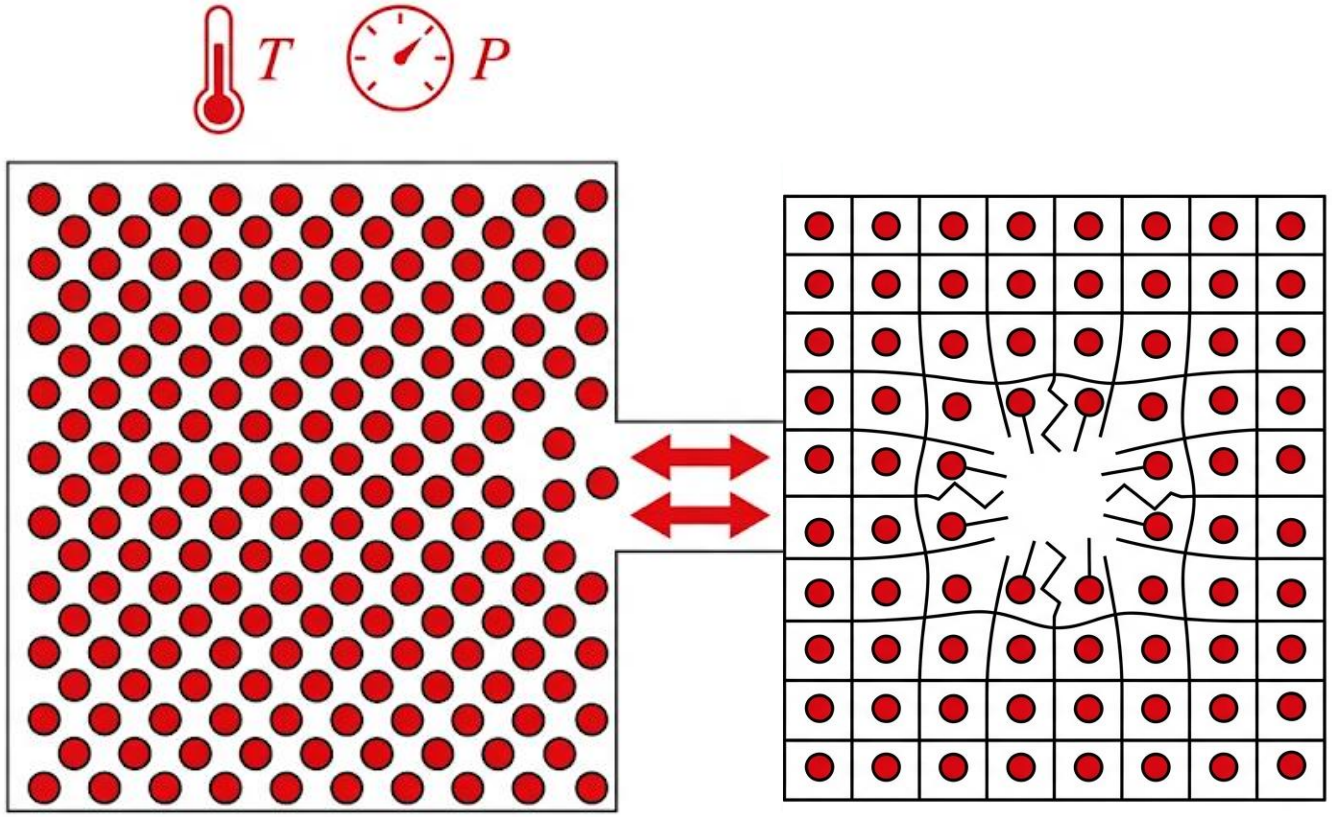
QUEEN'S UNIVERSITY BELFAST

The chemical potential μ_i of species i is the energy cost (or gain) of adding/removing one atom of that species to/from the crystal, **exchanged with a reservoir.**

$\Delta\mu_i = 0$: element-rich limit. More negative $\Delta\mu_i$: element-poor.



Convention: $\mu_i = \mu_i^o + \Delta\mu_i$, where μ_i^o is the reference chemical potential of the elemental phase (e.g. bulk metal, O_2 gas), and $\Delta\mu_i \leq 0$ is the deviation from that reference.



Chemical Potentials and Competing Phases: Binary (Al_2O_3)

QUEEN'S UNIVERSITY BELFAST

Stability condition:

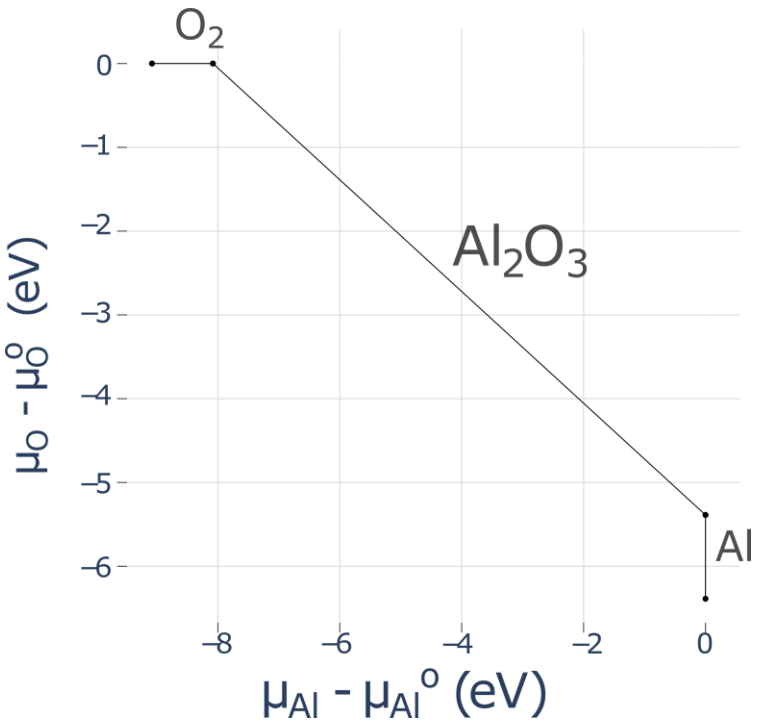
$$2\Delta\mu_{\mathrm{Al}} + 3\Delta\mu_{\mathrm{O}} = \Delta H_{\mathrm{f}}(\mathrm{Al_2O_3})$$

Al-rich bound:

$$\frac{1}{2}\Delta H_{\mathrm{f}}(\mathrm{Al_2O_3}) \leq \Delta\mu_{\mathrm{Al}} \leq 0 \text{ Al–rich}$$

O-rich bound:

$$O\text{–rich } 0 \geq \Delta\mu_{\mathrm{O}} \geq \frac{1}{3}\Delta H_{\mathrm{f}}(\mathrm{Al_2O_3})$$



Allowed (μ_{Al}, μ_O) line bounded by O_2 and Al elemental limits

Chemical Potentials and Competing Phases: Ternary ($CsPbI_3$)

QUEEN'S UNIVERSITY BELFAST

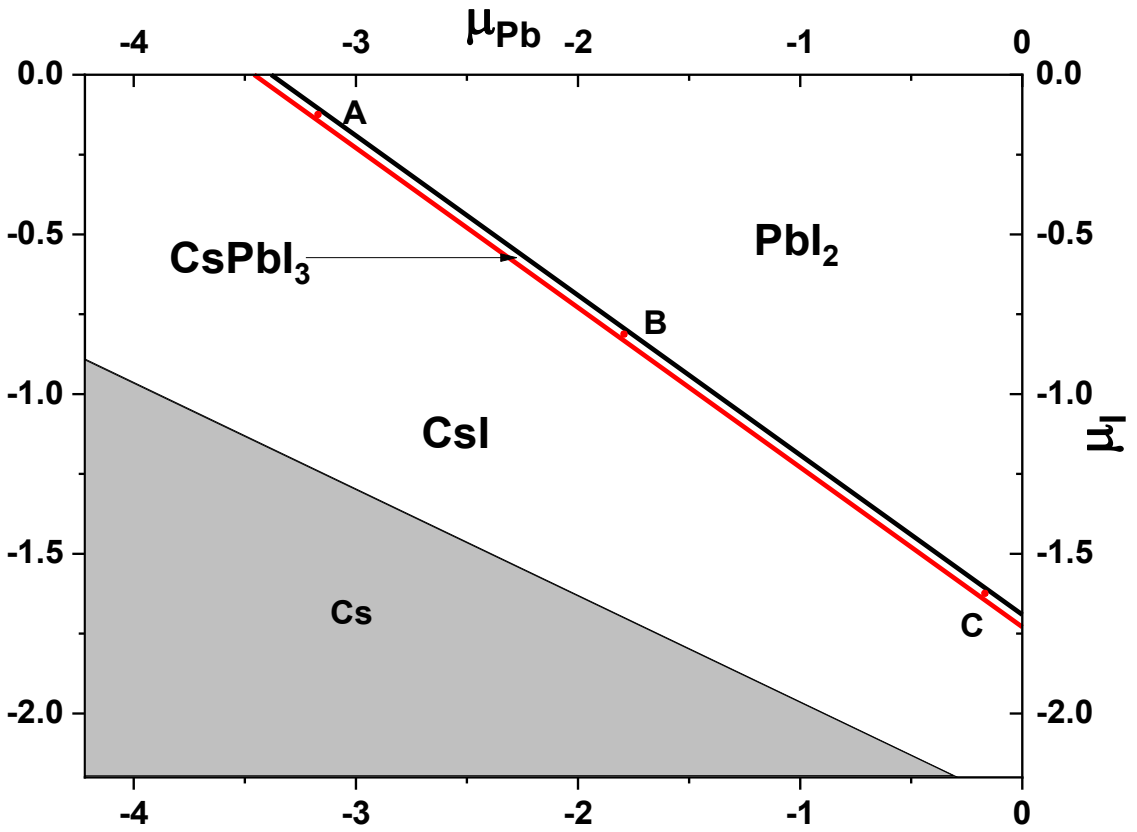
- Chemical reaction $CsI + PbI_2 = CsPbI_3$
- $Cs, Pb, I_2, CsI, PbI_2, CsPbI_3$

$$\Delta\mu_{Cs} + \Delta\mu_{Pb} + 3\Delta\mu_I = \Delta H = -5.88(\gamma CsPbI3)$$

$$\Delta\mu_{Cs} + \Delta\mu_I < \Delta H_{CsI} = -3.39\text{eV} \quad \text{(no CsI forms)}$$

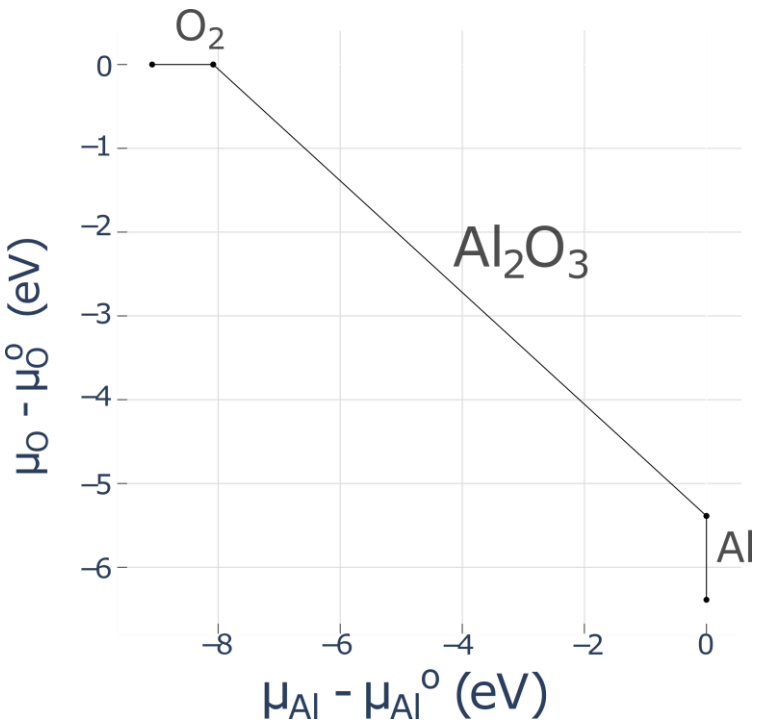
$$\Delta\mu_{Pb} + 2\Delta\mu_I < \Delta H_{PbI_2} = -2.40\text{eV} \text{ (no PbI2 forms)}$$

$$\Delta\mu_{Pb} + 2\Delta\mu_I > -2.49\text{eV}$$



−2.49 = −5.88 − (−3.39), so the stability window is 0.09 eV wide.

Chemical Potentials and Competing Phases: Extrinsic Dopants



Chemical potentials for Al under O-rich and O-poor conditions. Solubility-limiting secondary phases for dopants.

Si	Ge	Sn	C	Fe	Mg	N
Si	Ge	Sn	C_4	Fe	Mg	N2
Al_2SiO_5	$Al_2Ge_2O_7$	SnO_2	Al_4CO_4	Al_6Fe	$MgAl_2O_4$	AlN
SiO_2	GeO_2	SnO	CO_2	Fe_2O_3		N_2O_5

QUEEN'S UNIVERSITY BELFAST

Getting Competing Phases for a Doped System

Workflow:

1. Query a materials database (e.g. Materials Project) for all known phases in the relevant chemical space (host elements + dopant)
2. Retrieve formation energies / energies above hull for each candidate phase
3. Feed these into a chemical-potential-limits calculator

Tip: with a dopant the list grows fast (196 entries for Si – mostly SiO_2 polymorphs). Keep the e-above-hull tolerance tight.

In

```
from doped.chemical_potentials import CompetingPhases

cp = CompetingPhases("AlGaO3", energy_above_hull=0.03)  # queries MP
print(len(cp.entries))
print([entry.name for entry in cp.entries])
```

Out

```
18
['Al', 'Ga', 'O2', 'Al2O3', 'Ga2O3', 'Ga', 'Al2O3', 'Al', 'Ga',
 'Ga', 'Ga', 'Al2O3', 'Ga', 'AlGaO3', 'AlGaO3', 'Ga', 'Al2O3', 'Ga2O3']
```

In

```
cp_Si = CompetingPhases("AlGaO3", extrinsic="Si", energy_above_hull=0.03)
cp_Si.vasp_std_setup(user_incar_settings={"AEXX": 0.32, "HFSCREEN": 0.2,
                                          "ENCUT": 600})  # HSE-type inputs
```

QUEEN'S UNIVERSITY BELFAST

Filtering to the Phases That Actually Bound Your Window

Workflow:
Discard phases with energy well above the hull (thermodynamically irrelevant, won't form)
Keep only phases that share a facet with the host compound in composition space
For extrinsic dopants, identify the specific secondary phase that sets the solubility limit (as in slide 29's table)

Reading the output

- Two limits for a binary host: In-rich (In2O3–In) and O-rich (In2O3–O2).
- For each limit, the dopant μ is pushed as high as possible until a secondary phase forms: here SiO_2 sets the Si solubility limit.
- Result is stored in Si_cpa.chempots and is passed directly to the defect thermodynamics.

In

```
from doped.chemical_potentials import CompetingPhasesAnalyzer

# parse the DFT (vasprun.xml) results of the competing phases
Si_cpa = CompetingPhasesAnalyzer("In2O3", entries="Si/")
Si_cpa.calculate_chempots()
```

Out

```
Calculated chemical potential limits (in eV wrt elemental reference phases):

                In       O        Si  Si-Limiting Phase
Limit
In2O3-In  0.0000 -2.6959 -3.696798              SiO2
In2O3-O2 -4.0438  0.0000 -9.088598              SiO2
```

Plotting the Chemical Potential Diagram

QUEEN'S UNIVERSITY BELFAST

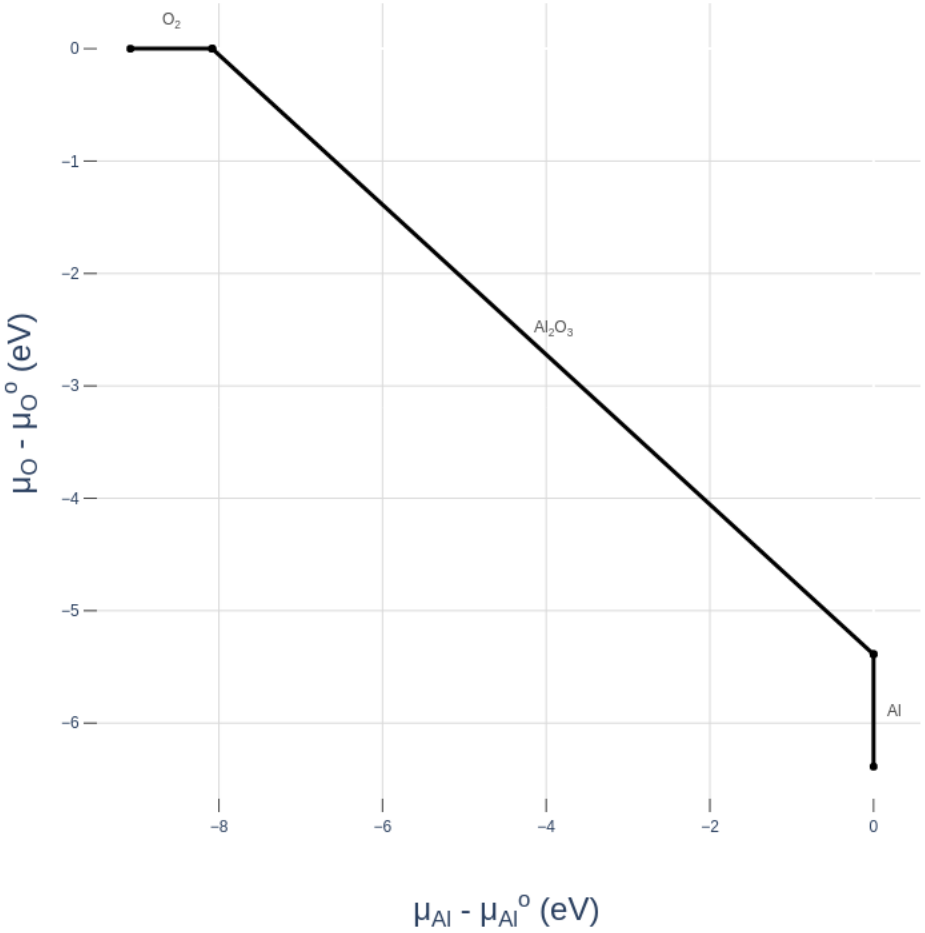
Reading the Output Diagram

```python
from pymatgen.analysis.chempot_diagram import ChemicalPotentialDiagram

# binary host (Al2O3): stability window is a line
cpd = ChemicalPotentialDiagram(cpa.intrinsic_phase_diagram.entries)
cpd.get_plot()

# ternary host (AlInO3): 2-D heat map, like slide 24
cpa.plot_chempot_heatmap(dependent_element="O")
```

```
              Al       O
Limit
Al2O3-Al  0.0000 -5.3874
Al2O3-O2 -8.0812  0.0000
```

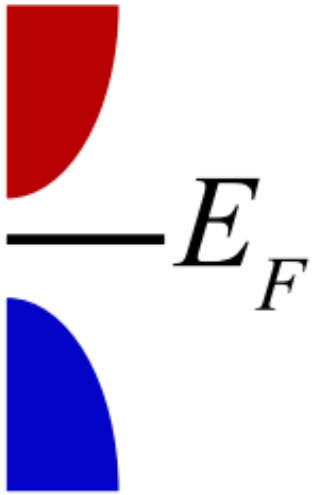


- Segment on the line = host is stable; end points = the Al-rich and O-rich limits.
- Check: $2\Delta\mu_{Al} + 3\Delta\mu_{O} = \Delta H_f(Al_2O_3) = -16.2$ eV per formula unit.
- This is exactly how the diagrams on slides 27–29 were generated.

The Electron Chemical Potential: The Fermi Level

QUEEN'S UNIVERSITY BELFAST

E_F is the chemical potential of electrons – the energy of the reservoir that supplies or absorbs electrons when a defect changes charge.



Where it enters: a defect in charge q has exchanged –q electrons with the reservoir, costing $q(E_{VBM} + E_F)$.

Allowed range: Usually plotted from VBM to CBM. With degenerate doping, E_F can move into the band

Slope = q: donors (q > 0) are cheap when E_F is low (p-type); acceptors (q < 0) are cheap when E_F is high (n-type).

Kinks = transition levels ε(q/q') – the E_F where two charge states cost the same (Lecture 3).

Possible Charge States of a Defect

QUEEN'S UNIVERSITY BELFAST

Workflow:

1 Start from oxidation states

- Al_2O_3: removing O^{2-} can leave up to 2 electrons → V_O ∈ {+2, +1, 0}; removing Al^{3+} → V_Al down to −3.
- doped guesses these automatically and adds a safety margin (e.g. V_O −1).

2 Adjust with physics / literature

- Remove unlikely extremes (e.g. Al_O +5).
- Add states suggested by experiment (EPR, DLTS, optical) or known polarons.

3 Confirm afterwards

- A charge state is only meaningful if it is stable for some E_F in the gap and the charge is localised.

In

```
from doped.generation import DefectsGenerator

defect_gen = DefectsGenerator(Al2O3_supercell,
                              generate_supercell=False,
                              interstitial_gen_kwargs=False)
```

Out

```
Vacancies            Guessed Charges     Wyckoff
-------------------  ------------------  -------
v_Al_Cs_O1.75        [+1,0,-1,-2,-3]     4i
v_O_Cs_Al1.75Al1.86  [+2,+1,0,-1]        4i
v_O_Cs_Al1.79        [+2,+1,0,-1]        4i

Substitutions         Guessed Charges     Wyckoff
--------------------  ------------------  -------
Al_O_Cs_Al1.79        [+5,+4,+3,+2,+1,0]  4i
O_Al_Cs_O1.75         [0,-1,-2,-3,-4,-5]  4i
```

In

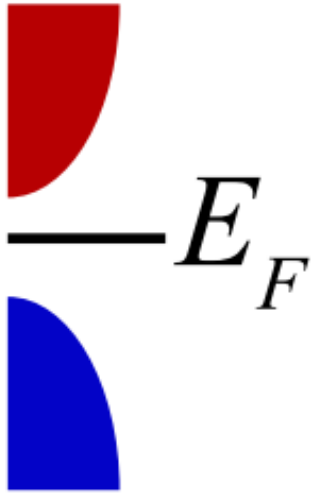
```
# trim or extend the guesses (literature / experiment)
defect_gen.remove_charge_states("Al_O_Cs_Al1.79", [+5, +4])
defect_gen.add_charge_states("v_O_Cs_Al1.79", [-2])
```

Fermi-Level Pinning

QUEEN'S UNIVERSITY BELFAST

E_F is not a free parameter: in equilibrium it is fixed **self-consistently** by overall charge neutrality.

$$\Sigma_q[X^q] + p(E_F) - n(E_F) = 0$$



- Every term depends on E_F → solve numerically (doped FermiSolver / py-sc-fermi).
- Pinning: once a compensating defect becomes cheap, adding more dopant just creates more compensators – E_F stops moving (dopability limit).

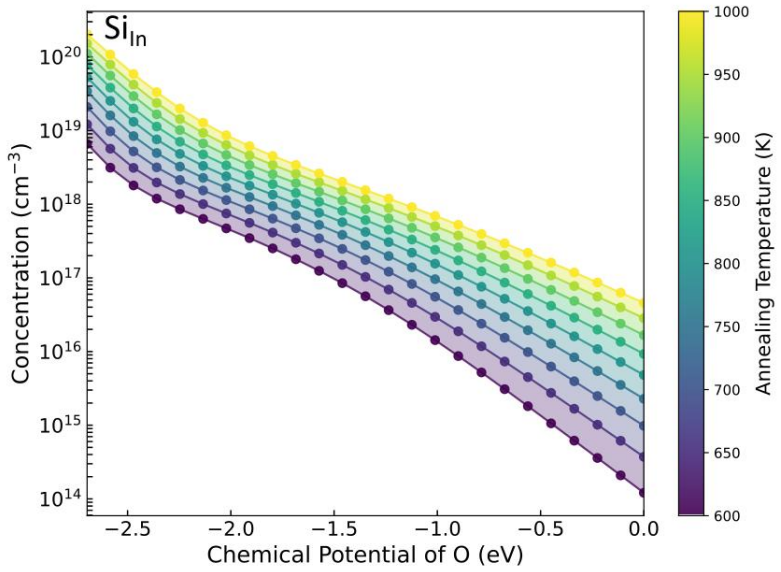
Practical Example: p-type vs n-type

QUEEN'S UNIVERSITY BELFAST

n-type: Si-doped In_2O_3 (O-rich)

```
from doped.thermodynamics import FermiSolver

fs = FermiSolver(defect_thermodynamics=defect_thermo,
                 bulk_dos="In2O3_dos/vasprun.xml",
                 backend="py-sc-fermi")
df = fs.scan_temperature(limit="O-rich",
                         temperature_range=range(300, 1410, 50))
```

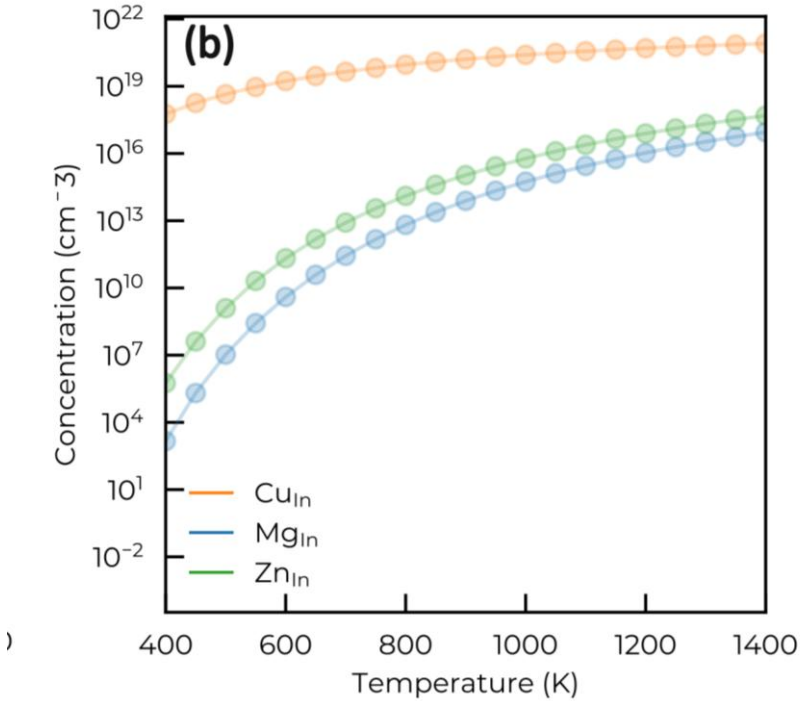


Fadla *et al.*, *PRMaterials* **9**, 105002 (2025)

Acceptors in In_2O_3

- Mg_In, Zn_In, Cu_In: (0/−) levels ≈ 1.5–1.7 eV above the VBM → deep acceptors, holes are not released.
- Their negative charge is compensated by native donors, so E_F stays far from the VBM.

Same workflow, opposite answer: In_2O_3 is easily n-type, practically not p-type.



Key Takeaways

QUEEN'S UNIVERSITY BELFAST

Reservoirs set the energy

Atoms exchange with μ_i, electrons with E_F.

A formation energy is meaningless without stating the limit (X-rich / X-poor) and E_F.

Competing phases define the allowed μ window.

Accuracy matters

0.1 eV ≈ 50× in concentration at 300 K.

Compute host, competing phases and defects with the same functional and settings.

Include corrections and entropy terms where they matter.

Reporting matters

Report μ limits and the competing phases used.

Band gap, VBM, dielectric constant, correction scheme.

Annealing / quenching T, code & version – so others can reproduce it.

Key Takeaways

QUEEN'S UNIVERSITY BELFAST

Quantity	Question
Chemical potential μ_i	What do atoms cost, and what sets that cost (growth conditions, competing phases)?
Competing phases	What thermodynamically limits the accessible range of μ_i?
Electron chemical potential (E_F)	What do electrons cost?
Fermi-level pinning	How is E_F actually fixed, self-consistently, by charge neutrality?
Carrier concentration	How sensitive is the predicted doping behaviour to errors in formation energy?

Lecture 3: Defect Formation Energy and Ionisation Levels

Mohamed Abdelilah Fadla

m.fadla@qub.ac.uk

Research fellow

School of Mathematics and Physics

Queen's University Belfast

Outline

QUEEN'S UNIVERSITY BELFAST

LECTURE 3

Defect Formation Energy and Ionisation Levels

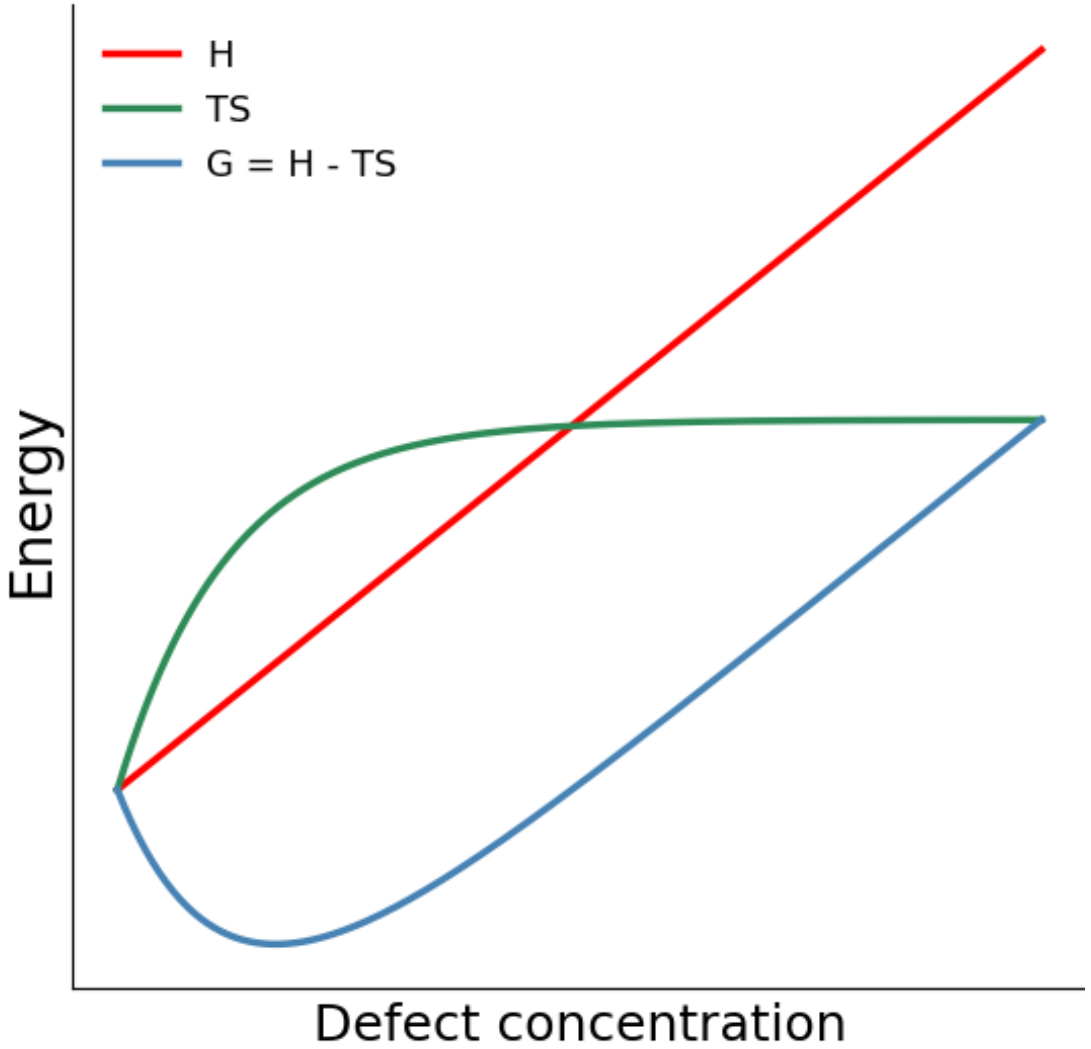
1. **Defect formation energy**
   Building the equation term by term
2. **Finite-size corrections**
   Spurious image interactions and the eFNV scheme
3. **Finding the right structure**
   Site screening and energy-lowering distortions (ShakeNBreak)
4. **Transition levels**
   Thermodynamic vs optical levels
5. **Guidance for reproducibility**
   DFT limitations and what to report

Recap: What We Covered

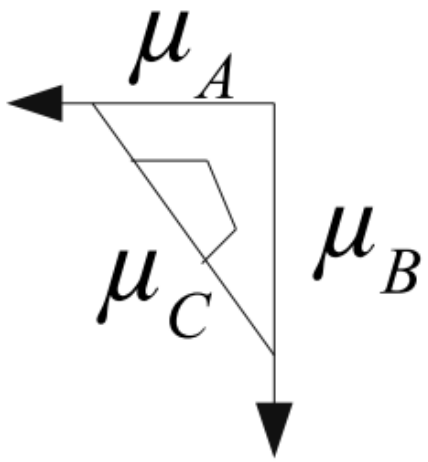
QUEEN'S UNIVERSITY BELFAST

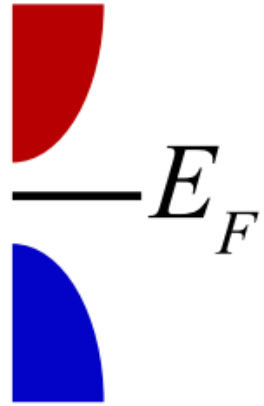
- **Enthalpy:** creating the defect costs energy
- **Entropy:** Many ways to place the defects; each defect can also change vibrations, spin and orientation.
- **Free Energy:** combines enthalpy and entropy to determine what the crystal actually does

$$c_{eq} = \exp\left(-\frac{\Delta g_f}{k_B T}\right)$$



Recap: What We Covered





Atom reservoir
μ_i (growth conditions)

⇄

Crystal + defect
$\Delta E_f(D, q)$

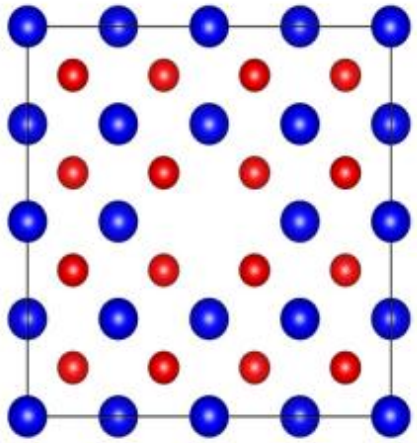
⇄

Electron reservoir
E_F (doping, carriers)

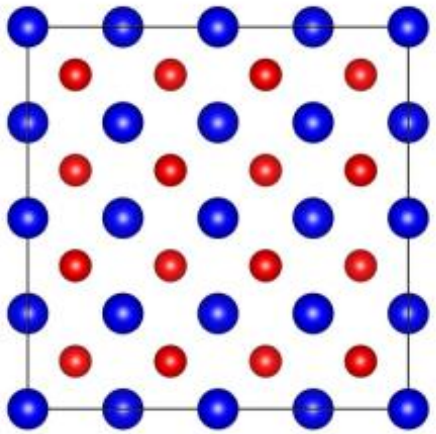
Defect Formation Energy

QUEEN'S UNIVERSITY BELFAST

The formation energy is the difference between the defective and pristine systems



-

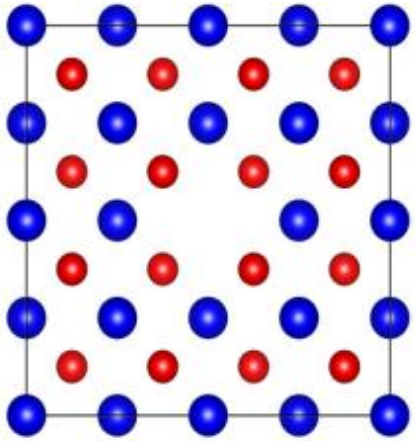


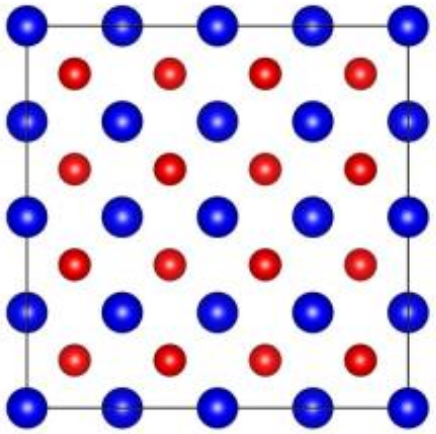
= ΔH_f

Defect formation energy

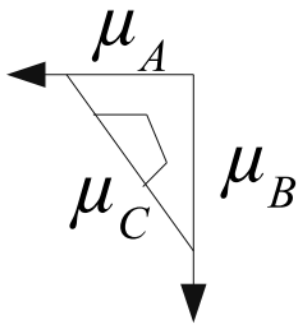
QUEEN'S UNIVERSITY BELFAST

$$E_{\text{tot}}\ [X_q] \qquad -E_{\text{tot}}\ [\ \text{bulk}\ ] \qquad -\sum n_i \mu_i$$





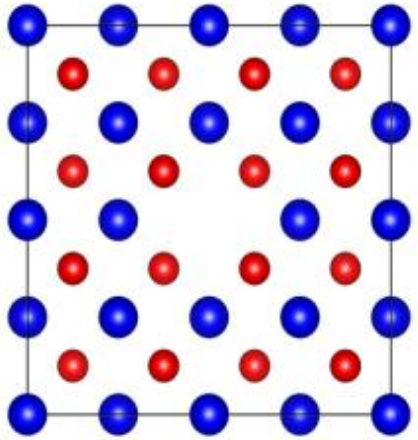
Chemical potentials



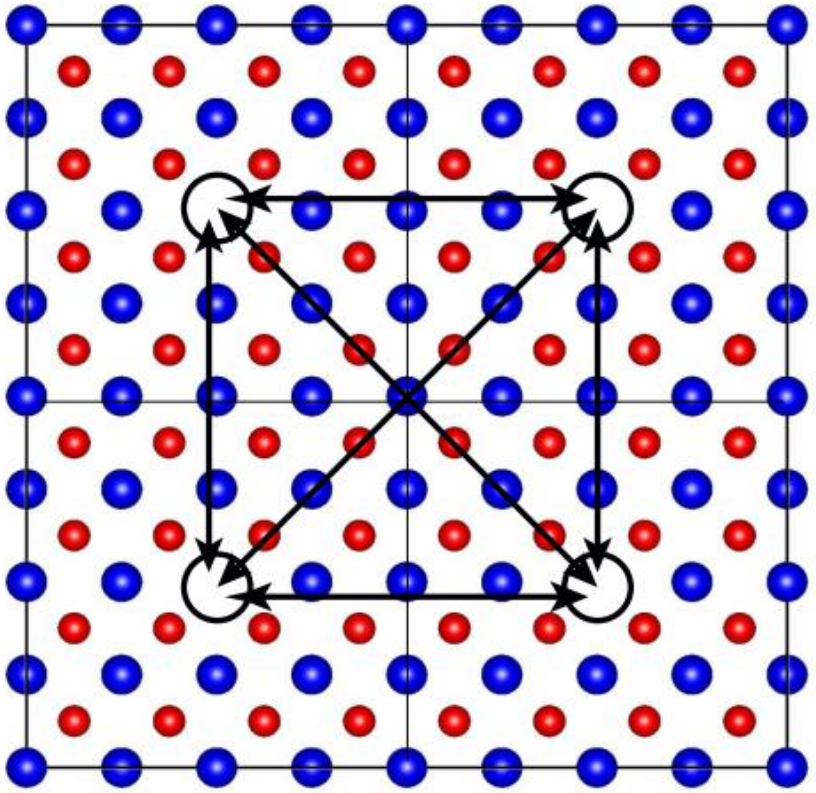
Defect formation energy

QUEEN'S UNIVERSITY BELFAST

What we want



What we model



We must know what we are modelling!

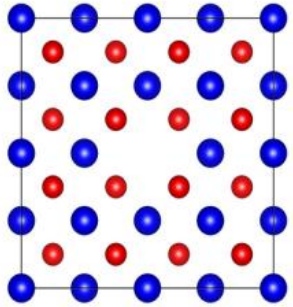
Valeria Ferrari's lecture (Monday)

Defect formation energy

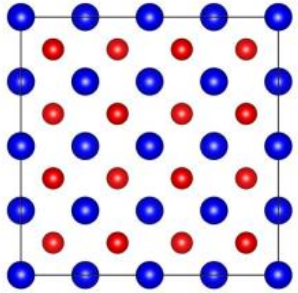
QUEEN'S UNIVERSITY BELFAST

$$E_{\text{tot}}\,[X_q] \quad - E_{\text{tot}}\,[\,\text{bulk}\,] \quad - \sum n_i \mu_i \quad + qE_{\text{F}} \quad + E_{\text{corr}}$$

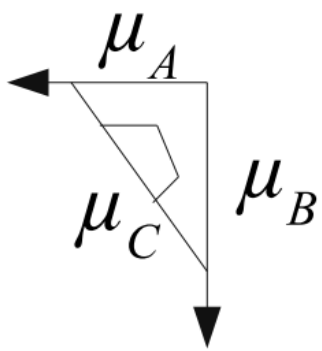
Defect Supercell



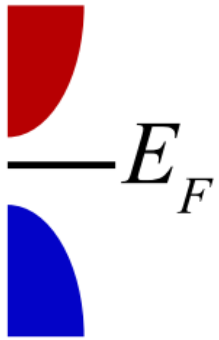
Host Supercell



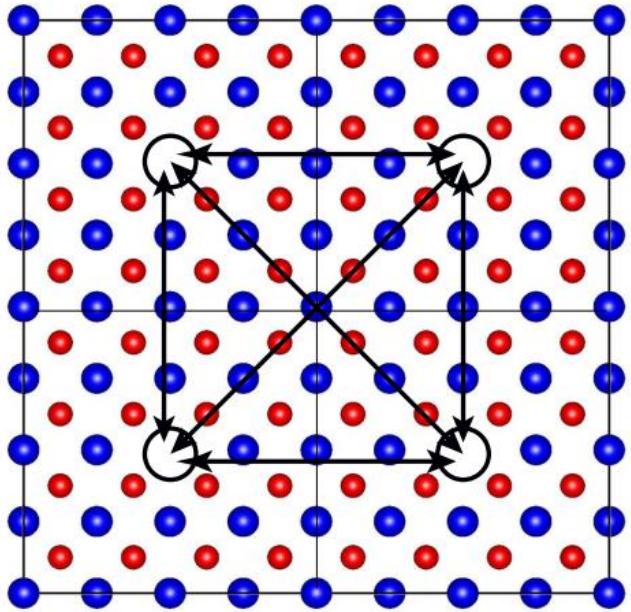
Chemical potentials



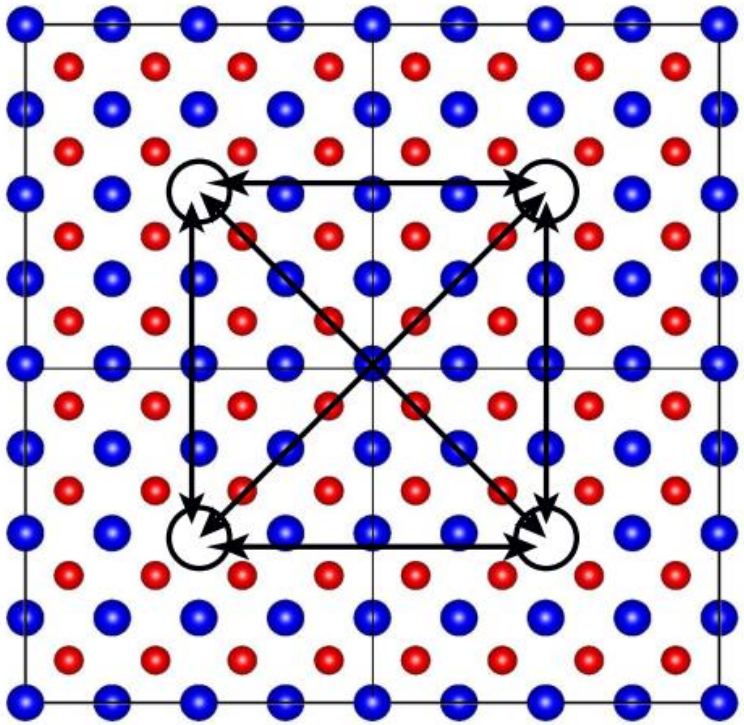
Electron chemical potential

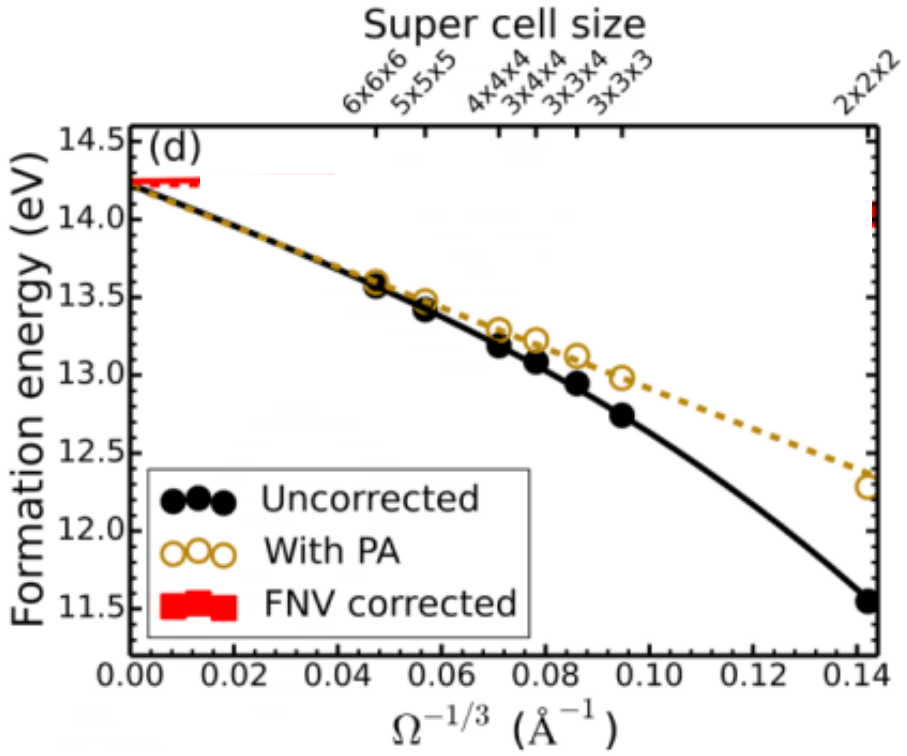


Finite Size correction



Defect formation energy and spurious interaction





Convergence of formation energies with respect to supercell size for uncorrected, with potential alignment (PA) only, and with the FNV scheme **M. H. Naik and M. Jain 2018**

eFNV Correction Scheme

QUEEN'S UNIVERSITY BELFAST

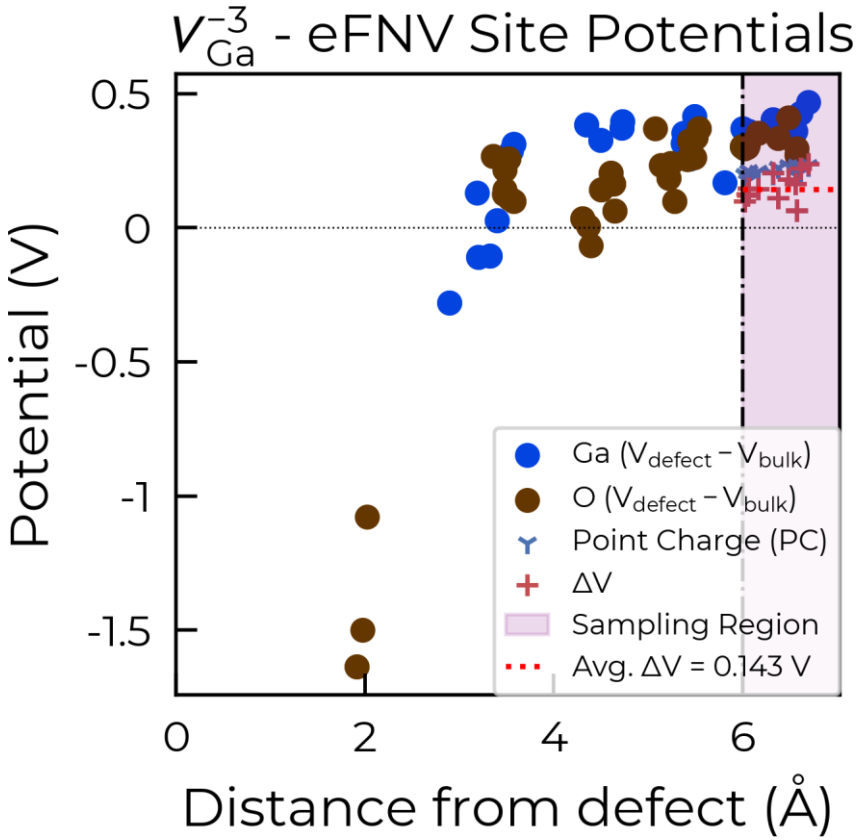
$$\Delta E_f(X^q) = E[X^q] - E[bulk] - \Sigma\, n_i\mu_i + q(E_{VBM} + E_F) + E_{corr}$$

image-charge term: a charged defect interacts with its periodic images and the neutralising background. Leading term ∝ $q^2/(\varepsilon L)$ → converges only as 1/L.

potential alignment: the average electrostatic potential shifts when the defect is added; aligned by comparing defect and bulk potentials far from the defect.

eFNV: uses atomic-site potentials and the full dielectric tensor → works for anisotropic, low-symmetry hosts.

Freysoldt, Neugebauer & Van de Walle, PRL 102, 016402 (2009); Kumagai & Oba, PRB 89, 195205 (2014)




```
dp = DefectsParser(".", bulk_path=bulk_path, dielectric=[8.2, 8.6, 9.1])
```

```
Cu_Al_-1: Charge = -1 with finite-size charge correction: +0.21 eV
Mg_Al_-1: Charge = -1 with finite-size charge correction: +0.23 eV
Zn_Al_-1: Charge = -1 with finite-size charge correction: +0.25 eV
```

Finding the Right Defect Configuration and Screening High-Energy Sites

QUEEN'S UNIVERSITY BELFAST

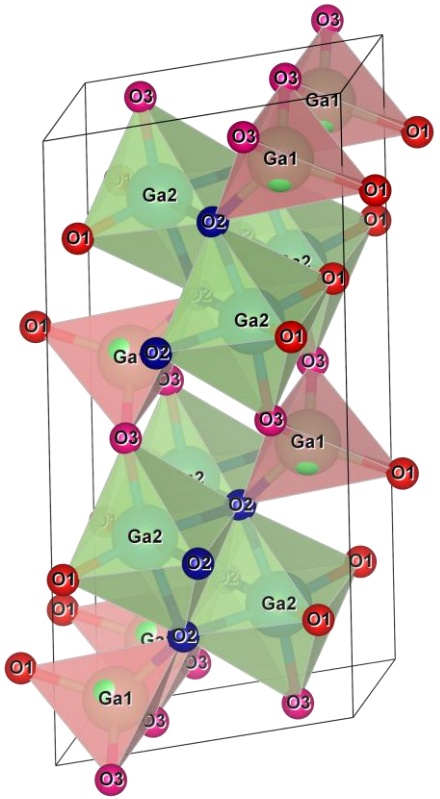
```
from doped.generation import DefectsGenerator
defects_generator = DefectsGenerator(host_struct_supercell,
  #extrinsic=["Mg","Zn","Cu","Hf","Zr","Ta","Sn","Si","C","Ge"],
                                     generate_supercell=False,
                                     #interstitial_gen_kwargs=False,
                                    )
```

Vacancies	Guessed Charges	Conv. Cell Coords	Wyckoff
v_Al_Cs_O1.75	[+1,0,-1,-2,-3]	[0.410,0.500,0.204]	4i
v_Al_Cs_O1.86	[+1,0,-1,-2,-3]	[0.158,0.500,0.318]	4i
v_O_Cs_Al1.75Al1.86	[+2,+1,0,-1]	[0.495,0.000,0.257]	4i
v_O_Cs_Al1.75Al1.87	[+2,+1,0,-1]	[0.161,0.000,0.109]	4i
v_O_Cs_Al1.79	[+2,+1,0,-1]	[0.173,0.000,0.567]	4i

Substitutions	Guessed Charges	Conv. Cell Coords	Wyckoff
Al_O_Cs_Al1.75Al1.86	[+5,+4,+3,+2,+1,0]	[0.495,0.000,0.257]	4i
Al_O_Cs_Al1.75Al1.87	[+5,+4,+3,+2,+1,0]	[0.161,0.000,0.109]	4i
Al_O_Cs_Al1.79	[+5,+4,+3,+2,+1,0]	[0.173,0.000,0.567]	4i
O_Al_Cs_O1.75	[0,-1,-2,-3,-4,-5]	[0.410,0.500,0.204]	4i
O_Al_Cs_O1.86	[0,-1,-2,-3,-4,-5]	[0.158,0.500,0.318]	4i

Interstitials	Guessed Charges	Conv. Cell Coords	Wyckoff
Al_i_C2	[+3,+2,+1,0]	[0.000,0.144,0.000]	4g
Al_i_C2h	[+3,+2,+1,0]	[0.500,0.500,0.500]	2c
Al_i_Cs_Al1.66	[+3,+2,+1,0]	[0.097,0.500,0.019]	4i
Al_i_Cs_Al1.68	[+3,+2,+1,0]	[0.398,0.000,0.434]	4i

.
.
.

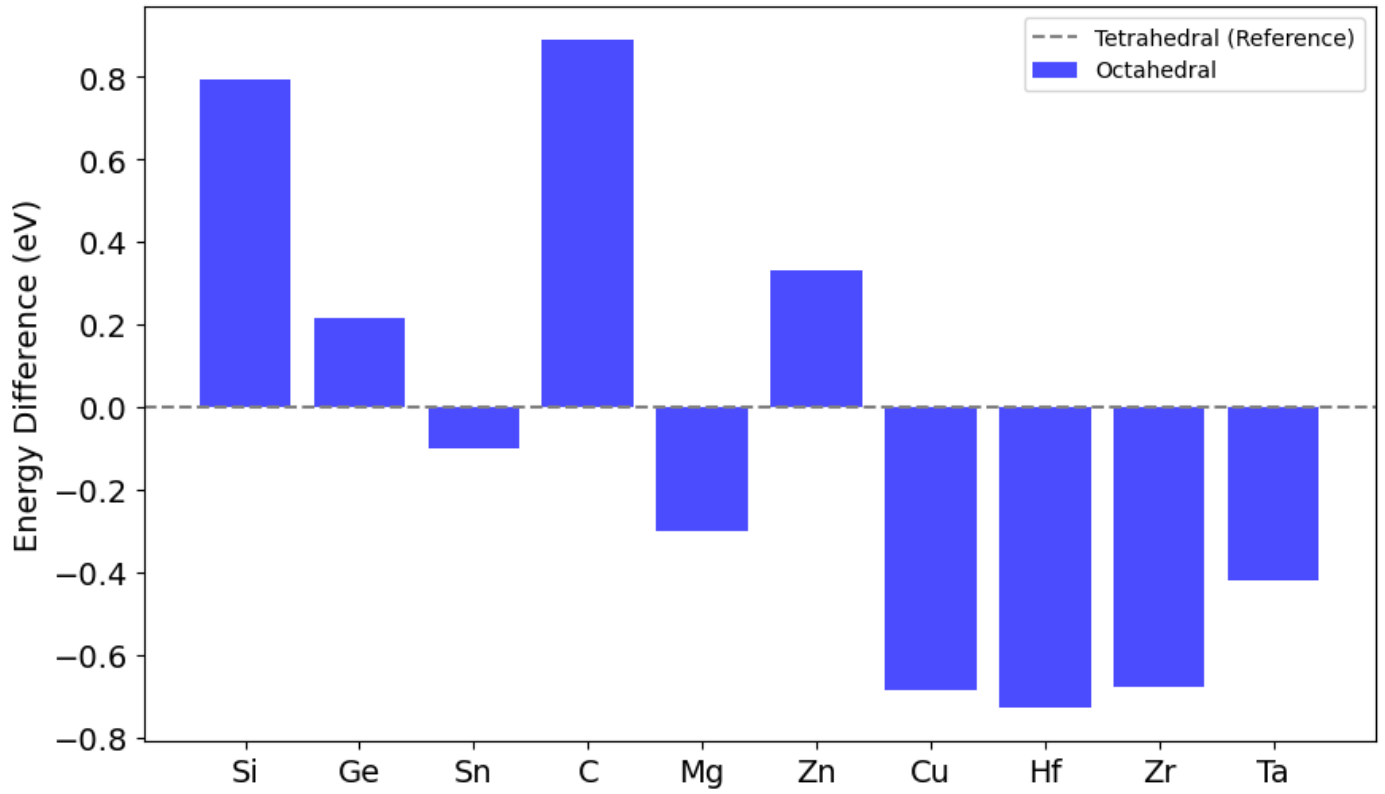


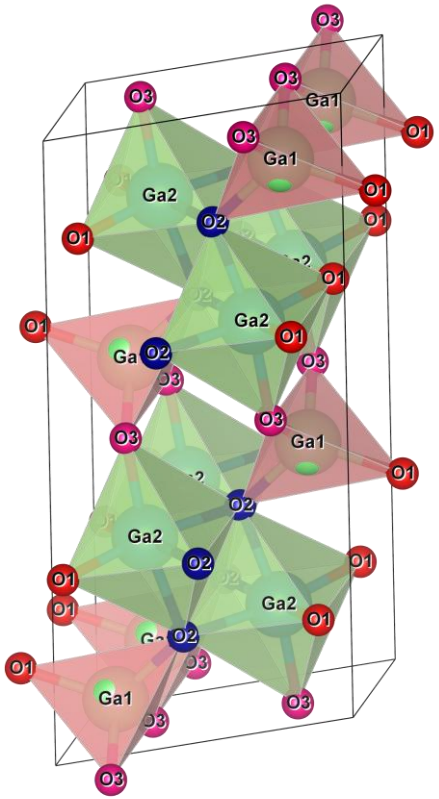
Low symmetry!

Finding the Right Defect Configuration and Screening High-Energy Sites

QUEEN'S UNIVERSITY BELFAST

Screen out high-energy sites to focus on the most relevant defect configurations.



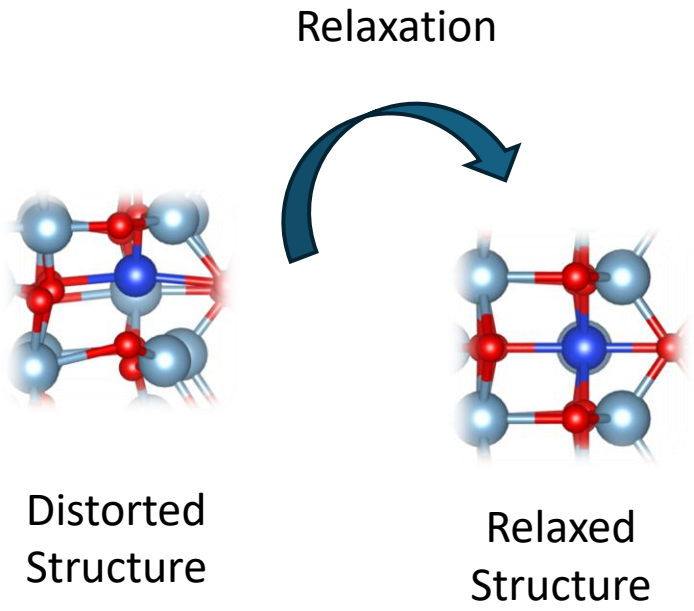


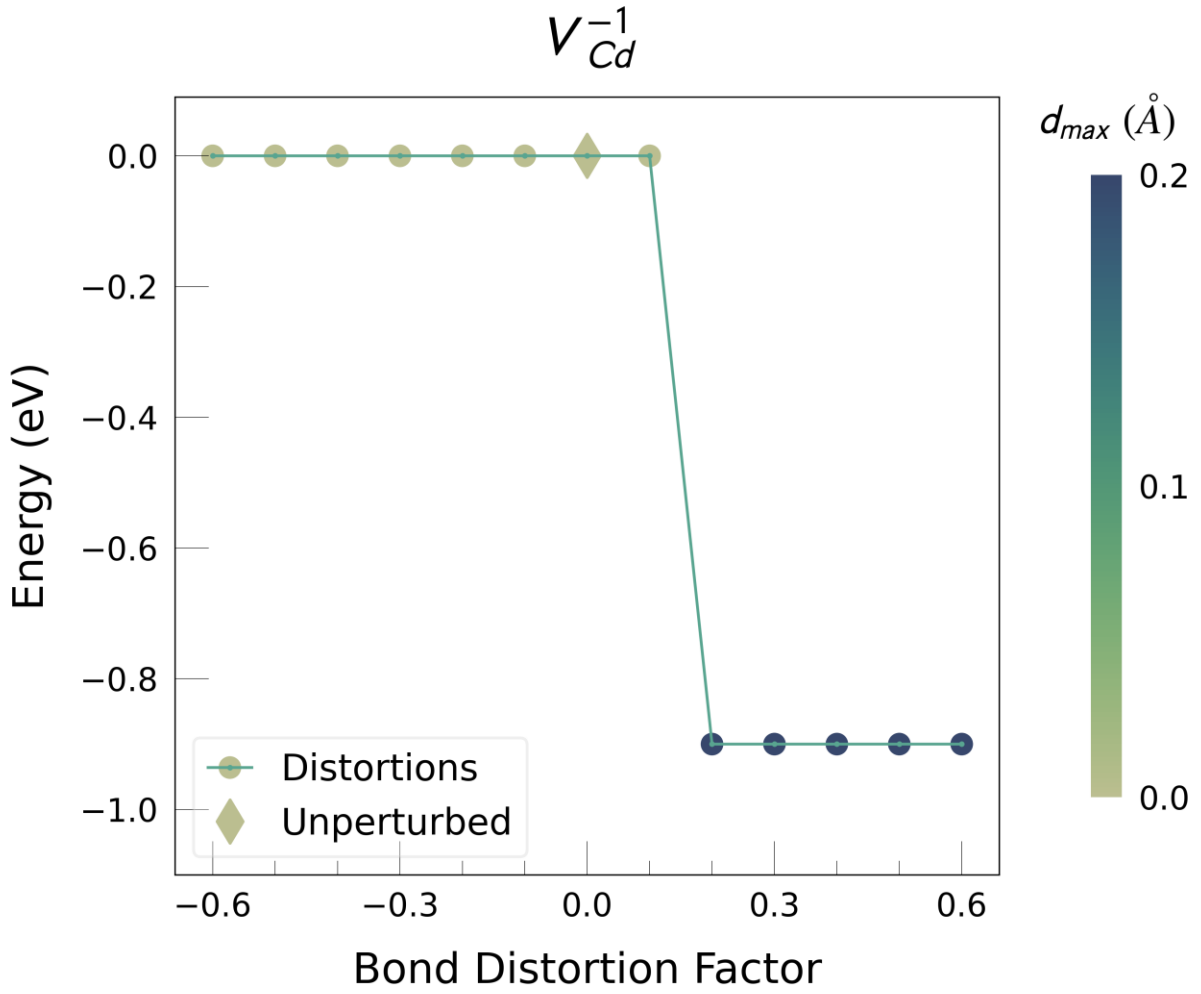
Low symmetry!

Energy-lowering distortions

QUEEN'S UNIVERSITY BELFAST

Identify ground state structures of defects by generating distorted defect structures and looking for any energy-lowering distortions.





Check: shakenbreak.readthedocs.io

Metastable configuration: ShakeNBreak

QUEEN'S UNIVERSITY BELFAST

Identify ground state structures of defects by generating distorted defect structures and looking for any energy-lowering distortions.

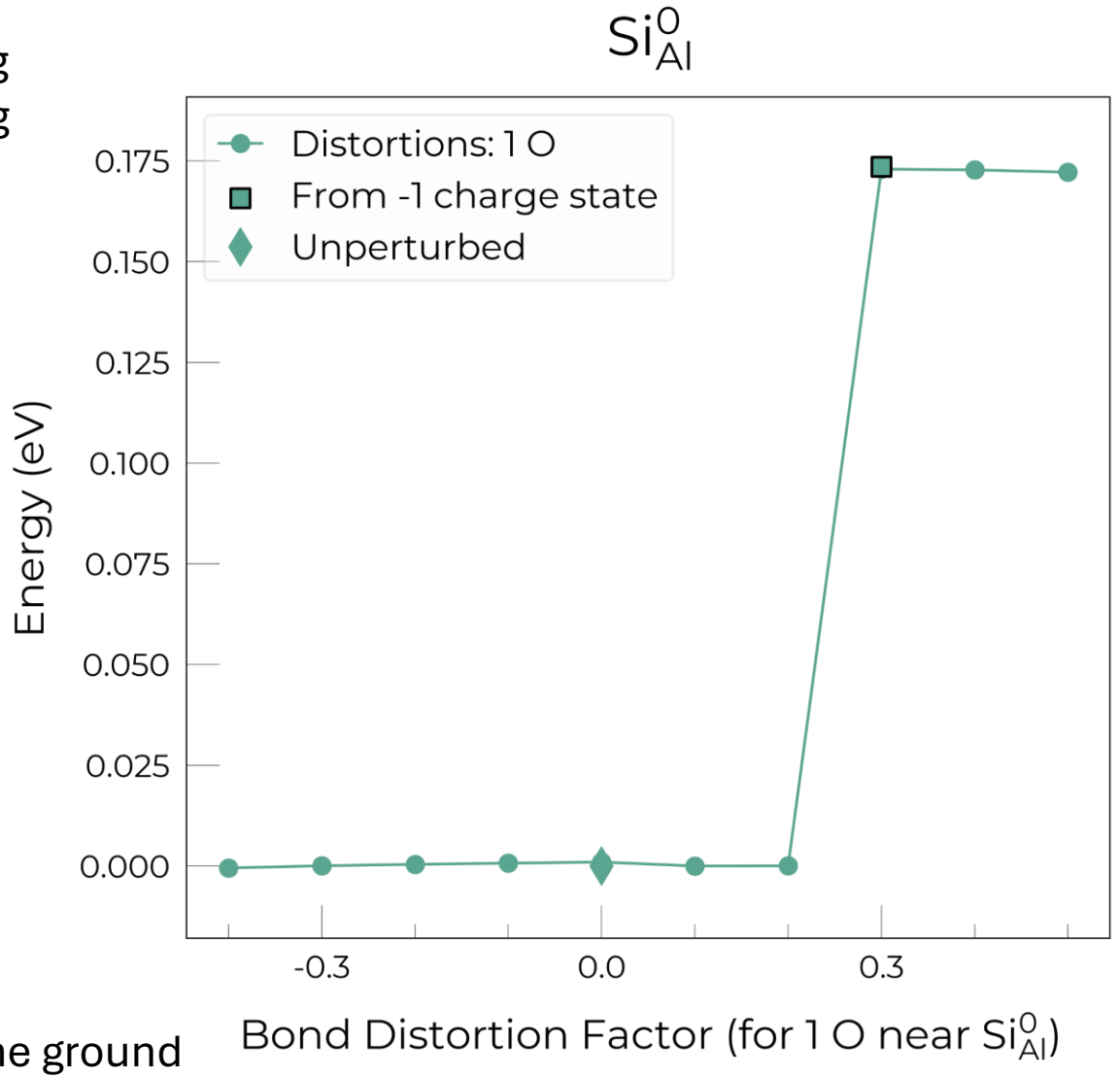
```python
from shakenbreak.input import Distortions

Dist = Distortions(defect_entries=defect_gen)
defects_dict, metadata = Dist.write_vasp_files(
    user_incar_settings={"AEXX": 0.32, "HFSCREEN": 0.2, "ENCUT": 600})
```

```
Applying ShakeNBreak... Will apply the following bond distortions:
['-0.6', '-0.5', ..., '0.5', '0.6']. Then, will rattle with a std dev of 0.18 Å

Defect: v_Ga_Cs_O1.83
Number of missing electrons in neutral state: 3
Defect v_Ga_Cs_O1.83 in charge state: +1. Number of distorted neighbours: 4
Defect v_Ga_Cs_O1.83 in charge state: -1. Number of distorted neighbours: 2
```

```
snb-run --submit-command sbatch --job-script job.sh   # then snb-parse, snb-plot
```



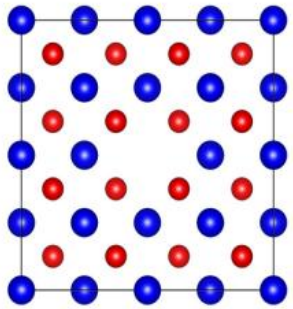
Si_Al^0: no lowering found, so the unperturbed structure is the ground state. Metastability; Walter Meyer's lecture (yesterday)

Defect formation energy

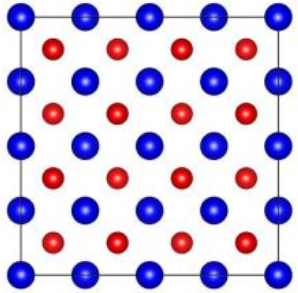
QUEEN'S UNIVERSITY BELFAST

$$E_{\text{tot}}\,[X_q] \quad - E_{\text{tot}}\,[\,\text{bulk}\,] \quad - \sum n_i \mu_i \quad + qE_{\text{F}} \quad + E_{\text{corr}}$$

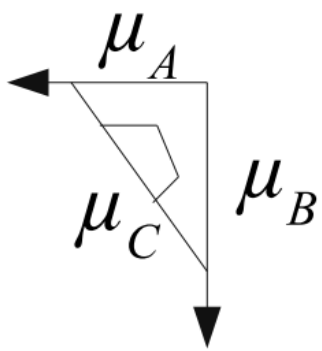
Defect Supercell



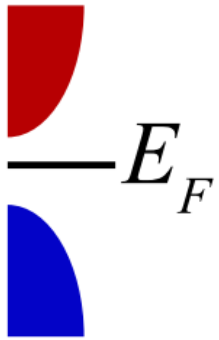
Host Supercell



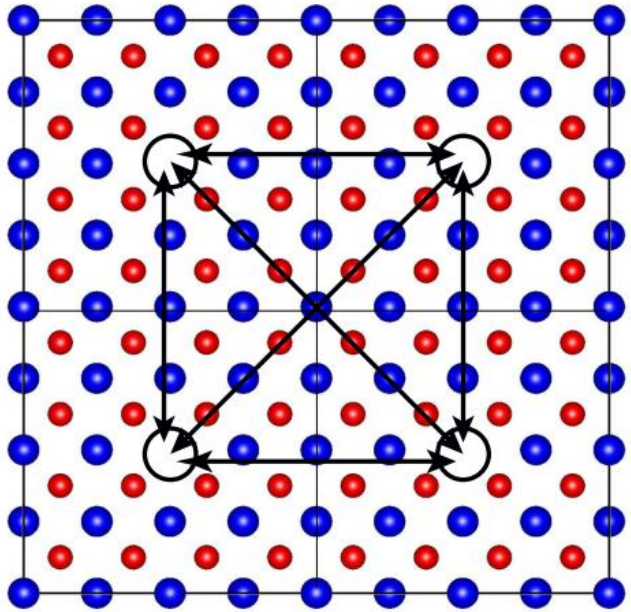
Chemical potentials



Electron chemical potential



Finite Size correction

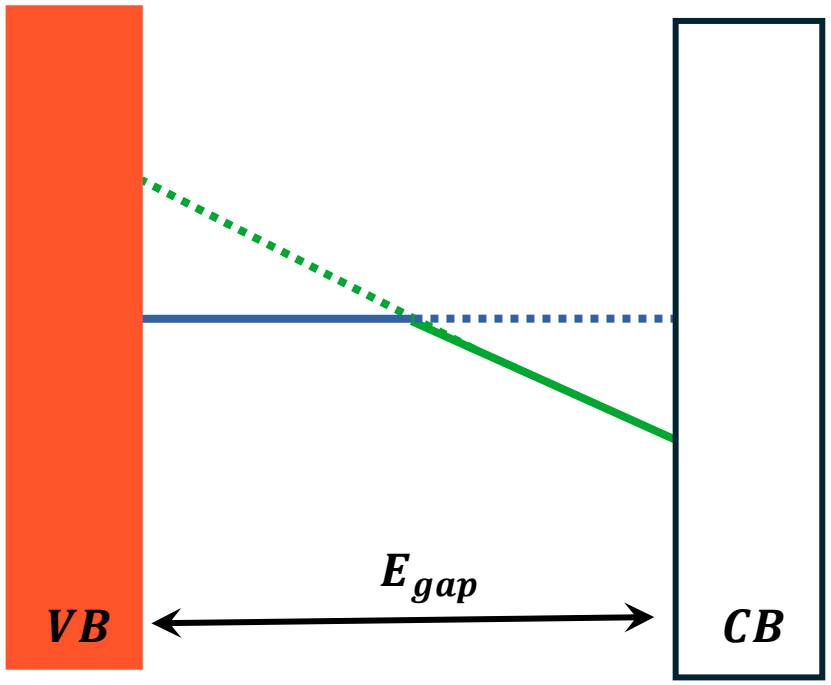


We have everything!

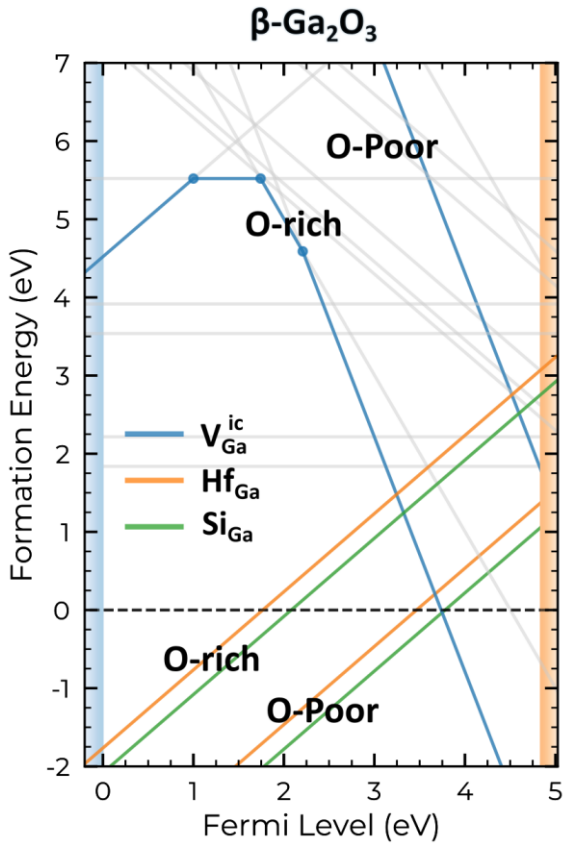
The formation energy of a defect in charge state q depends linearly on the Fermi level

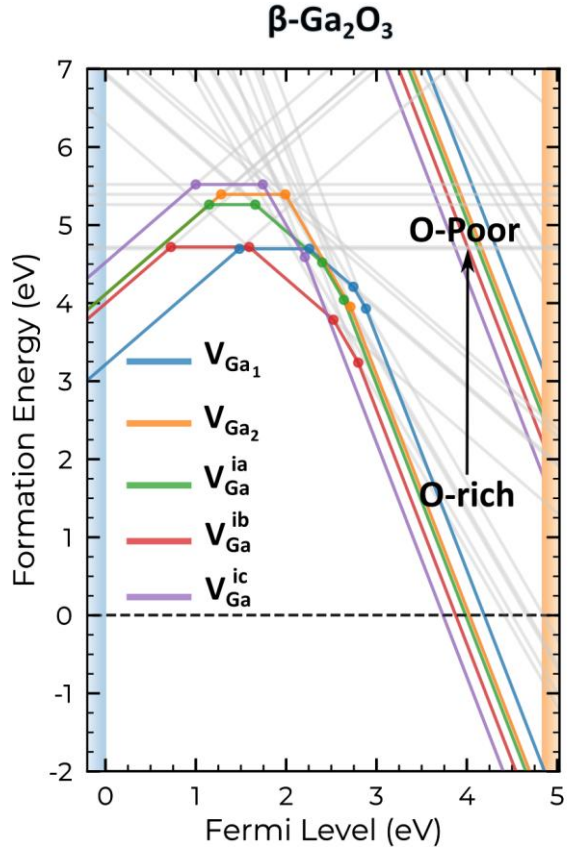
Defect formation energy

$$\Delta E_f(X^q) = E[X^q] - E[bulk] - \Sigma\, n_i\mu_i + \boldsymbol{q(E_{VBM} + E_F)} + E_{corr}$$



Formation energy diagram

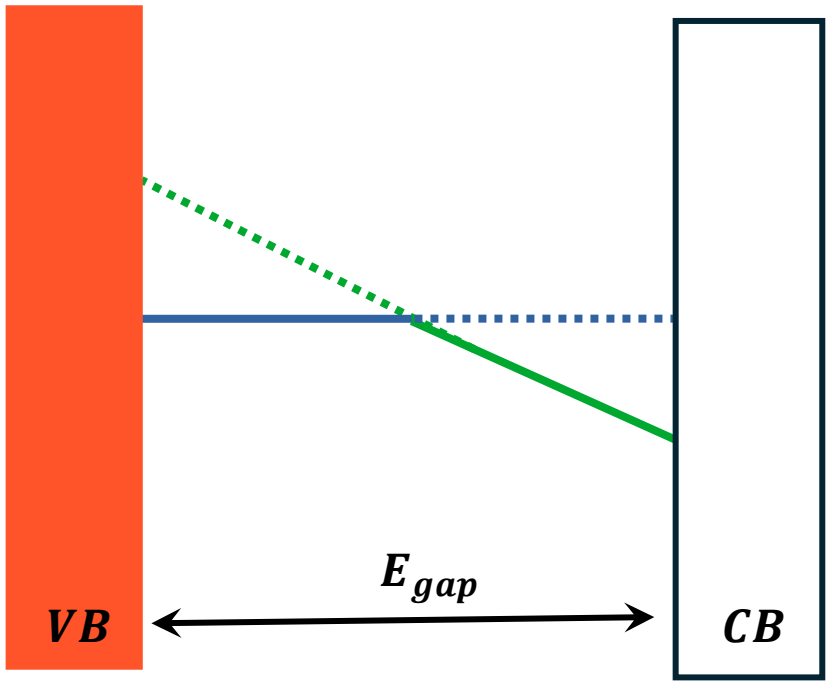




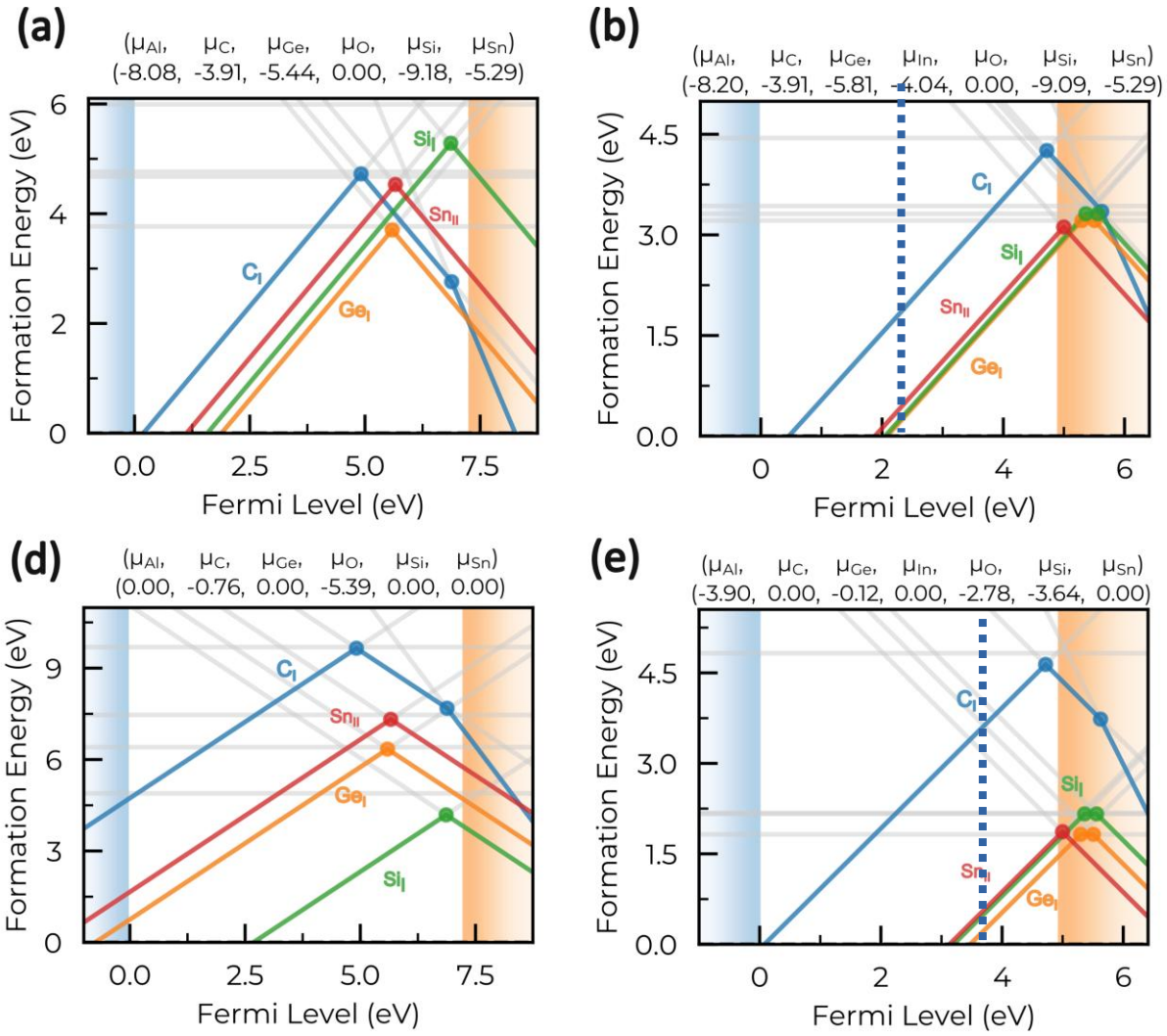
Defect formation energy

QUEEN'S UNIVERSITY BELFAST

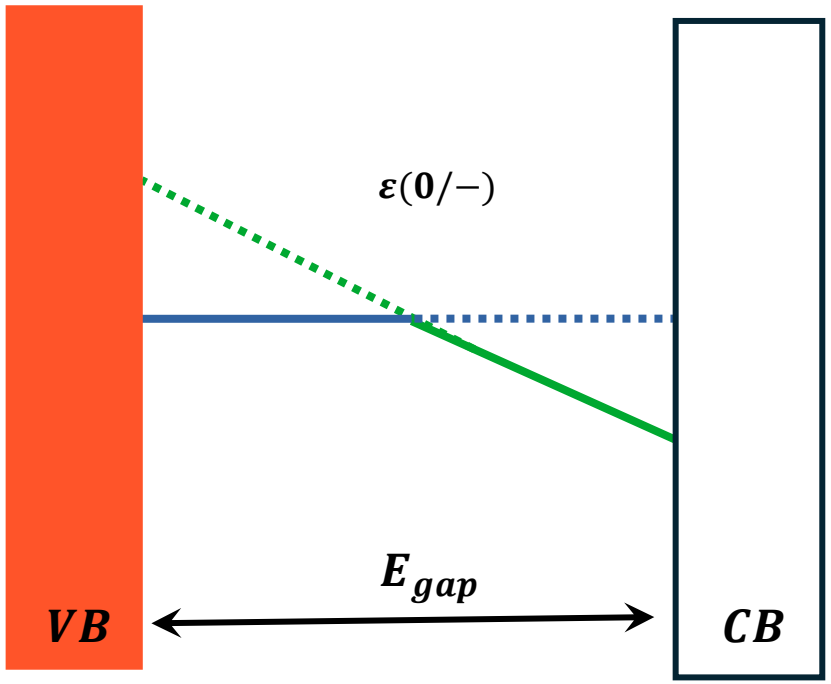
$$\Delta E_f(X^q) = E[X^q] - E[bulk] - \Sigma\, n_i\mu_i + \boldsymbol{q(E_{VBM} + E_F)} + E_{corr}$$

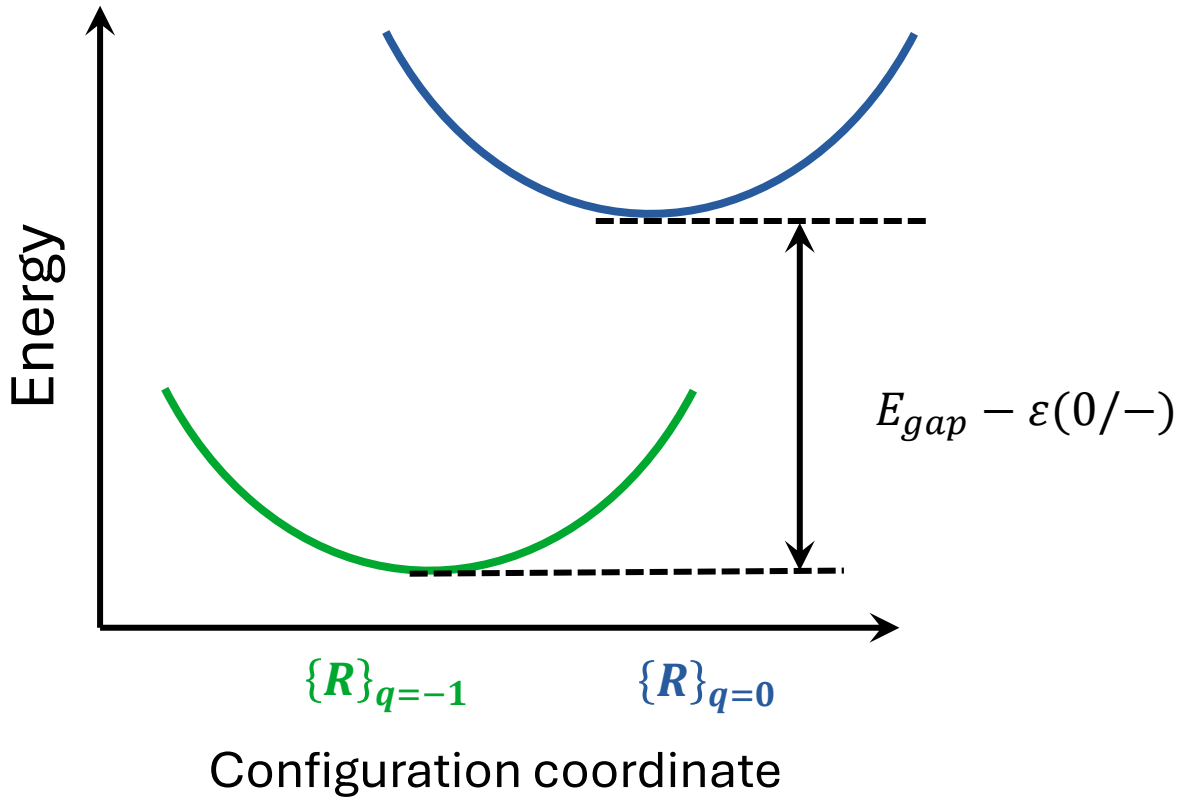


Formation energy diagram

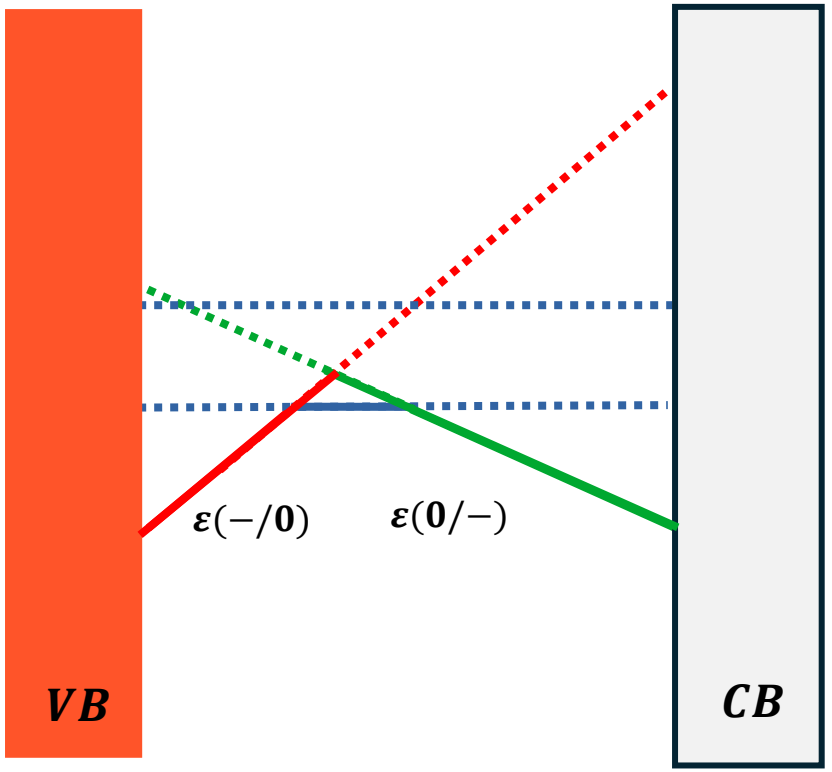


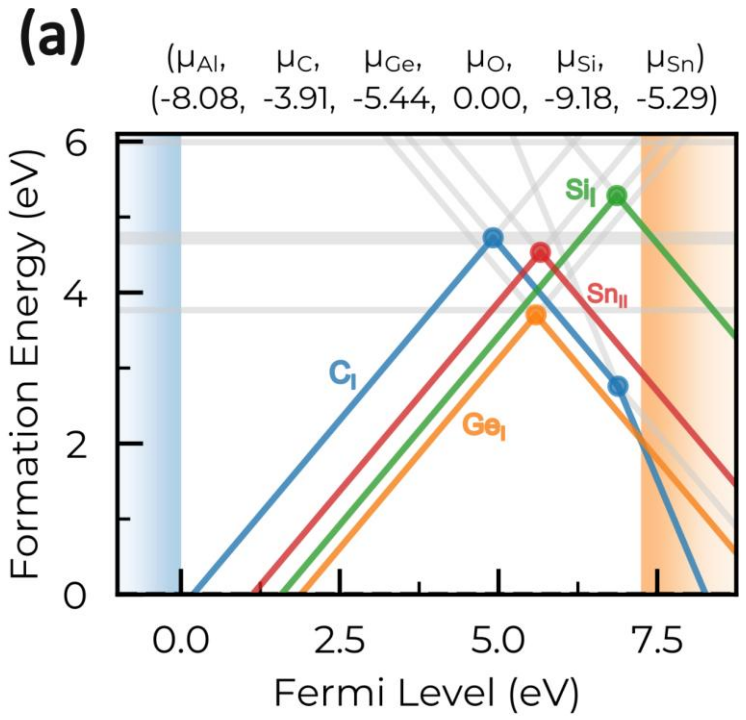
Thermodynamic transition levels

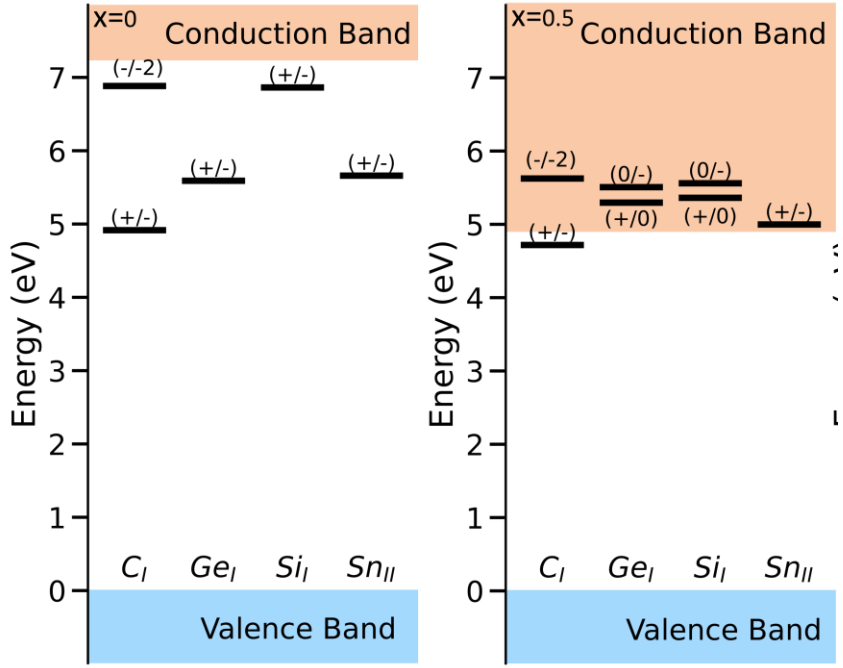




Thermodynamic transition levels





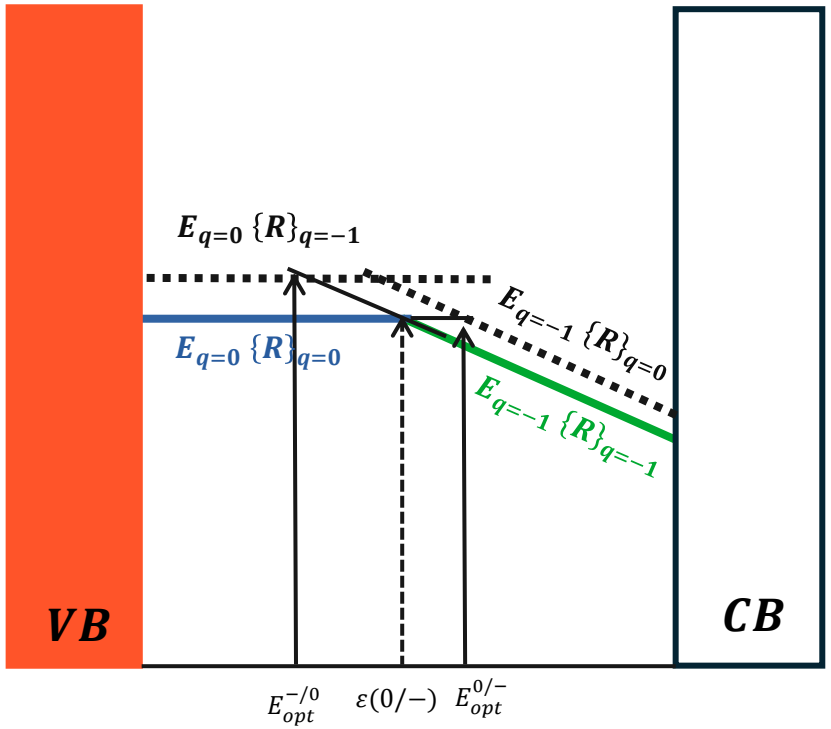


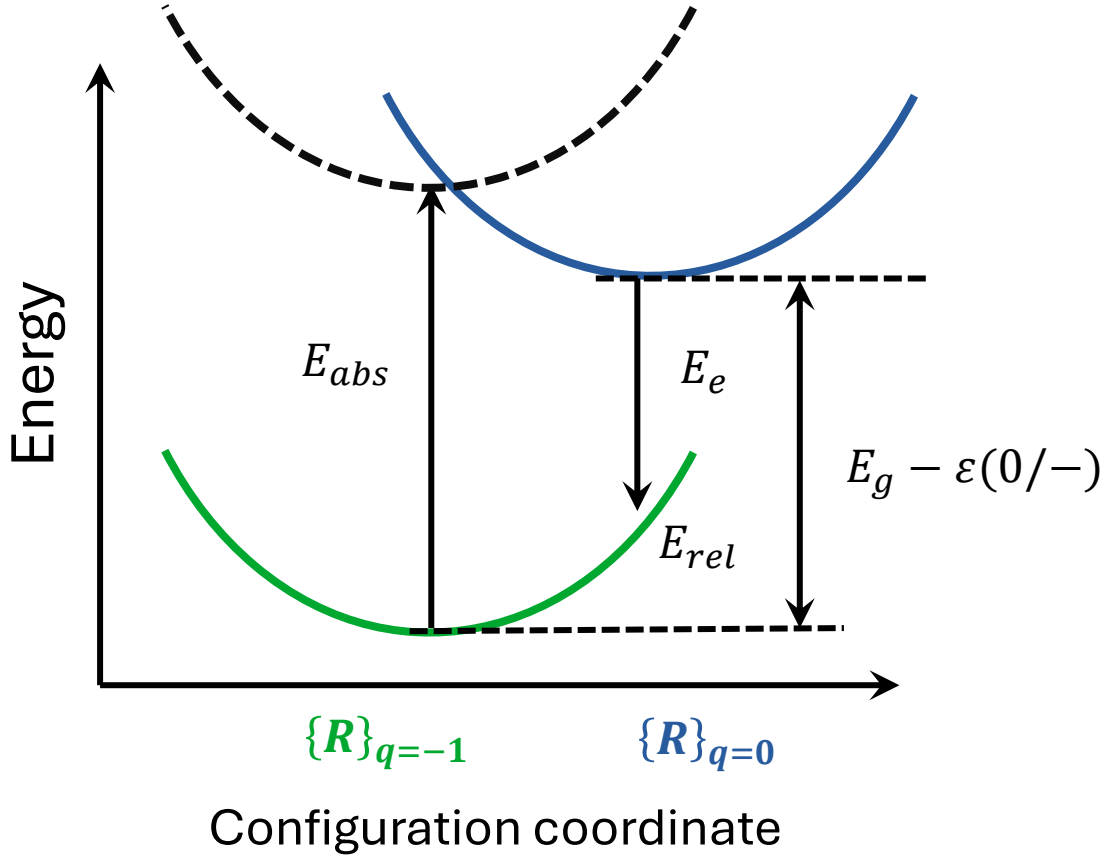
Negative U

Walter Meyer's lecture (yesterday)

Shallow vs deep defects:

Optical transition levels





Transition	Process	Geometry held fixed	Energy
E_{abs}	$D^- + hv \rightarrow D^0 + e^-(CBM)$	that of q = −1	$E_g - E_{opt}^{-/0}$
E_e	$D^0 + e^-(CBM) \rightarrow D^- + hv$	that of q = 0	$E_g - E_{opt}^{0/-}$
ZPL	relaxed → relaxed	—	$E_g - \varepsilon(0/-)$

DFT limitations

QUEEN'S UNIVERSITY BELFAST

Self-interaction error in semi-local DFT spreads defect-bound electrons/holes over many atoms: localised states, polarons and deep levels are missed, and band gaps are underestimated.

Level of theory	Band gap	Charge localisation / defect geometry
PBE (GGA)	Strongly underestimated	Often delocalised; wrong ground-state configuration common
SCAN (meta-GGA)	Improved, still too small	Better in some cases, but can still fail to localise defect/polaron states
HSE06 / tuned hybrid	Close to experiment (α can be tuned)	Generally localises correctly; standard for defect levels
GW / beyond	Quasiparticle accuracy	Band edges & levels; usually on top of hybrid geometries

- In our oxides, both PBE and SCAN gave the wrong defect configuration for many defects → all production runs use a hybrid (AEXX = 0.32 tuned to the band gap).
- Practical compromise: screen structures cheaply (MLFF), then relax the lowest candidates with the hybrid.

What to report

QUEEN'S UNIVERSITY BELFAST

- ✓ **Electronic structure**
  code & version, functional (mixing α), pseudopotentials, cut-off, k-mesh
- ✓ **Supercell**
  size / shape, minimum image distance, relaxation thresholds
- ✓ **Corrections**
  scheme (eFNV …), dielectric tensor used, spin polarisation
- ✓ **Chemical potentials**
  competing phases considered, limits used for each plot
- ✓ **Thermodynamics**
  band gap & VBM, DOS, annealing / quenching T, entropy terms

Further reading

Freysoldt et al., *Rev. Mod. Phys.* **86, 253 (2014)**

Squires et al., Nat. Rev. Mater. 2026, and Mosquera-Lois et al. 2023

– first-principles point-defect calculations: the standard reference

Tools

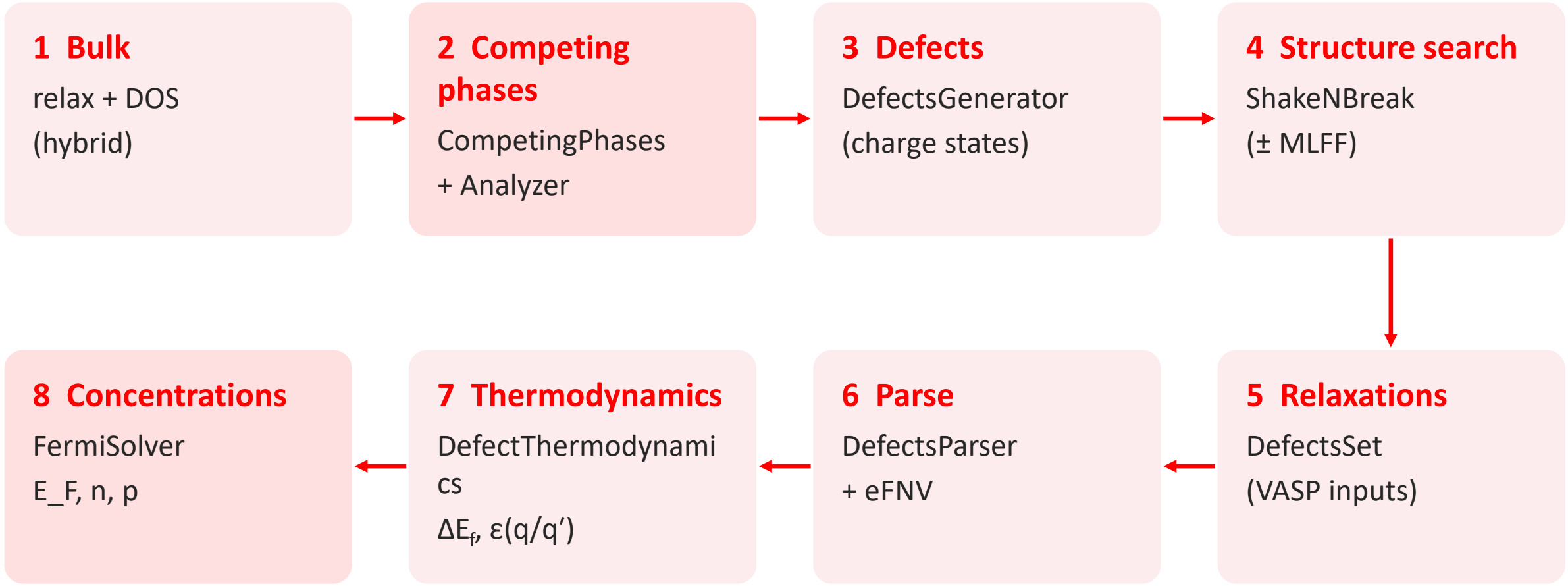
doped · ShakeNBreak · py-sc-fermi · pymatgen-analysis-defects

Share inputs & outputs

e.g. on Zenodo / NOMAD, so results are reproducible.

Full workflow

QUEEN'S UNIVERSITY BELFAST



Loop back to step 3–4 if new low-energy structures or charge states appear.

¡Gracias!

¿Preguntas? · Questions?

m.fadla@qub.ac.uk

MAMBA International School 2026 · INAOE, Puebla

Concentration

$$\frac{\partial G_{d,P}}{\partial n_d} = \frac{\partial\left(G_{b,P} + G_{f,P} - TS^{conf}\right)}{\partial n_d} = \frac{\partial\left(G_{f,P} - TS^{conf}\right)}{\partial n_d}$$

$$\frac{\partial G_{d,P}}{\partial n_d} = \frac{\partial\left(n_d g_{f,P} - TS^{conf}\right)}{\partial n_d}$$

$$c_{eq} = \frac{n_d}{N} = \exp\left(\frac{-g_{f,P}}{k_B T}\right)$$

$$S^{conf} = k_B \ln(W)$$

$$g_{f,P} = h_{f,P} - Ts_{f,P}$$

$$W = {}^{N}C_{n_d} = \frac{(N)!}{(N-n_d)!n_d!} \approx \frac{(N)^{n_d}}{n_d!}$$

$$c_{eq} = \exp\left(\frac{s_{f,P}}{k_B}\right)\exp\left(\frac{-h_{f,P}}{k_B T}\right) = \left(\frac{Z_d}{Z_b}\right)\exp\left(\frac{-h_{f,P}}{k_B T}\right)$$

$$S^{conf} = k_B[n_d - n_d \ln(n_d/N)]$$

Quantity	Formula	Units
Site fraction	$c = n_d/N = \exp(-g_f / k_BT)$	dimensionless (per lattice site)
Number of defects	$n_d = N \cdot \exp(-g_f / k_BT)$	count
Concentration per volume	$n_d/V = (N/V) \cdot \exp(-g_f / k_BT)$	cm^{-3}